COS30049 Computing Technology Innovation Project

Assignment 2 delivery plan

Everything that will be built, measured, written and packaged between 20 October and 8 November 2026

Project	Bitcoin Address Activity Explorer (unusual-activity indicator, never a fraud verdict)
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Plan prepared	Tuesday 20 October 2026 (teaching Week 6, day 36 of the 84-day unit)
Assignment 2 due	Sunday 8 November 2026, 23:59 GMT+7 (day 55); 19 days from this plan
Planned submission	Saturday 7 November 2026, one day early
Training machine	Intel Core i5-12600K, NVIDIA RTX 3060 12 GB, 16 GB DDR4-3600, Windows

What this document is. A working plan for finishing every marked part of Assignment 2: the datasets, the code, the trained models (including two language models trained from scratch on the machine above), the README and the 3,000-word report. It says what will be done, in what order, with which commands, how long each step should take, how each step is checked, and which rubric line it earns.

What it is not. It is not the report and is not submitted. Numbers marked est. are planning estimates that a measured run replaces. Nothing in here is a result.

Out of scope here. The weekly meeting minutes and the contribution form are also submitted with Assignment 2. The team will handle them separately; Section 15.4 only lists them so they are not forgotten.

Source documents read for this plan: Week 1–8 lecture slides, seminars and workshop notebooks; the submitted Assignment 1 report, minutes and contribution form; the Assignment 1 working files (handover audit, risk register, fix log); the Assignment 2 brief and rubric; the prototype repository (commit 8375cf0); and the existing `asm2/` language-model code.

Contents

1 The plan on one page **4**

2 What is submitted and how it is marked **5**
2.1 The files that go to Canvas . . . 5
2.2 Where each point is earned . . . 5
2.3 Hard constraints from the brief . . . 7
2.4 Promises Assignment 1 made about Assignment 2 . . . 8

3 Starting point **9**
3.1 What already exists . . . 10
3.2 What is missing . . . 10
3.3 What the unit's own material says to do . . . 10

4 Key decisions **11**
4.1 D-A2-1: one model family for each type the brief names . . . 11
4.2 D-A2-2: datasets . . . 12
4.3 D-A2-3: where accuracy, precision, recall and F1 are allowed . . . 14
4.4 D-A2-4: two language models trained from scratch . . . 14
4.5 D-A2-5: one repository and two environments . . . 15
4.6 D-A2-6: change control . . . 15

5 The machine and the compute budget **15**
5.1 What the hardware can do . . . 16
5.2 How long the text model's pretraining takes . . . 16
5.3 Decision gate on pretraining length . . . 16
5.4 Compute budget per job . . . 16
5.5 Running four lanes at once . . . 17
5.6 Windows housekeeping before a multi-day run . . . 17

6 Workstream 1: data collection **18**
6.1 DS1: the core snapshot . . . 18
6.2 DS2: the sequence corpus for TxGPT . . . 19
6.3 DS3: Elliptic . . . 19
6.4 DS4: BitcoinHeist . . . 20
6.5 DS5 and DS6: text corpora . . . 20
6.6 DS7: the explanation set . . . 20
6.7 Collection problems and responses, for the report . . . 21

7 Workstream 2: processing and feature engineering **21**
7.1 DS1: from raw pages to a clean transaction table . . . 21
7.2 Features F1–F4: a port of the prototype . . . 22
7.3 The time split . . . 22
7.4 Synthetic injection: known positives in real histories . . . 22
7.5 DS3 Elliptic processing . . . 23
7.6 DS4 BitcoinHeist processing . . . 23
7.7 DS2: TxGPT tokenisation . . . 23
7.8 Leakage controls in one place . . . 24
7.9 Tests . . . 25

8 Workstream 3: data analysis **25**
8.1 DS1 and DS2 analyses . . . 26
8.2 DS3 and DS4 analyses . . . 27

8.3 What the report's analysis section will contain . . . 27

9 Workstream 4: the scikit-learn models 27
9.1 Selection criteria (the report's model-selection section) . . . 27
9.2 Candidates considered . . . 28
9.3 Baseline: robust z (A1 §2.3) . . . 29
9.4 Clustering: K-Means and DBSCAN . . . 29
9.5 Regression: expected behaviour and surprise . . . 30
9.6 Classification on DS3 and DS4 . . . 30
9.7 Isolation Forest and LOF (the A1 extras) . . . 31
9.8 Search budget and saved artefacts . . . 32

10 Workstream 4b: the two language models 32
10.1 TxGPT: a transaction-sequence model . . . 32
10.2 The Explorer Assistant: a 152M text model . . . 33
10.3 The Explorer Assistant 3B: a fine-tuned comparison (added 22 October) . . . 34
10.4 GPU timetable . . . 35

11 Workstream 5: evaluation 35
11.1 On our own data (DS1): the A1 label-free plan, kept and extended . . . 35
11.2 On the labelled data (DS3, DS4) . . . 36
11.3 Language models . . . 36
11.4 Choosing the method the Explorer uses . . . 37
11.5 Comparing results against the objectives . . . 37
11.6 Unexpected outcomes: planned for, not hoped against . . . 37

12 Workstreams 6–9: prediction, code quality, README, packaging 38
12.1 Prediction interface . . . 38
12.2 Code structure and comments . . . 38
12.3 README . . . 39
12.4 Datasets and packaging . . . 40

13 Workstream 10: the report 41
13.1 Cover and front matter (not counted) . . . 41
13.2 Section-by-section plan . . . 42
13.3 Appendices (not counted) . . . 43
13.4 Writing rules . . . 44
13.5 Referencing . . . 44
13.6 Word count . . . 44
13.7 How the report gets produced . . . 44
13.8 Completeness checklist . . . 45

14 Schedule 45
14.1 Day by day . . . 45
14.2 Gantt chart . . . 47
14.3 Milestones and what each one proves . . . 47
14.4 The critical path and the slack . . . 47

15 Risks, priorities and the submission checklist 48
15.1 Risk register for Assignment 2 . . . 48
15.2 Early-warning signals . . . 51
15.3 Priority list . . . 52
15.4 Submission checklist . . . 52
15.5 After submission: what Assignment 3 inherits . . . 53

A Change request CR-06 and the messages to send **54**
A.1 Change log entry . . . 54
A.2 Effect on the A1 work breakdown . . . 54
A.3 Approval steps . . . 55
A.4 Message to the tutor . . . 55
A.5 Message to the team . . . 56

B Datasets considered **56**

C Draft reference list **58**

D Command reference, in run order **62**
D.1 Day 1 (21 October): set-up and checks . . . 62
D.2 Data collection (Section 6) . . . 62
D.3 GPU lane (Sections 5.3 and 10) . . . 62
D.4 Processing (Section 7) . . . 63
D.5 Analysis and models (Sections 8 and 9) . . . 63
D.6 Evaluation and outputs (Section 11) . . . 63
D.7 Prediction (Section 12.1) . . . 63
D.8 Report checks and packaging (Sections 13 and 15.4) . . . 64

1 The plan on one page

Goal. A full-marks attempt at all 40 points: 12 for the report, 20 for the code and zip file, and 8 HD points. Every task in this plan is tied to one of those points. If a task earns no point and nothing else depends on it, it was left out.

The marks, checked against the brief. The breakdown used to start this plan missed one report line. The report is worth 12 points over *six* criteria, not five:

Report (PDF, up to 3,000 words)	**Pts**	**Code and zip file**	**Pts**
1. Data collection and processing	5	1. Core functionality	10
2. Data analysis	2	2. Datasets	4
3. Justification of model selection	2	3. README (conda set-up, training, prediction)	2
4. Writing quality and logical structure	1	4. Comments	2
5. Harvard referencing	1	5. Code structure	2
6. Completeness (every required section)	**1**		
Report total	*12*	*Code total*	*20*
HD: additional datasets		4	
HD: additional ML implementations		4	

What will be built.

1. **Our own Bitcoin data.** A frozen snapshot of 25 addresses plus 3 short-history ones from the Esplora API, as Assignment 1 promised (DS1). A second, much larger crawl of about 5,000 address histories (DS2) feeds the transaction language model.
2. **Two public labelled datasets.** Elliptic (DS3) and BitcoinHeist (DS4) carry labels from their authors, so the brief's accuracy, precision, recall and F1 can be reported honestly.
3. **All three model types the brief names.** Clustering (K-Means, DBSCAN) and regression (Ridge, random forest, gradient boosting) run on our own data. Classification (logistic regression, KNN, random forest, gradient boosting) runs on the labelled datasets. The brief needs two types; doing all three removes any doubt.
4. **The Assignment 1 methods, kept as extra experiments.** The robust-z baseline, Isolation Forest and Local Outlier Factor are still built and evaluated with the label-free plan. They are also tested against the Elliptic labels, which shows how often "unusual" and "labelled illicit" coincide.
5. **Two language models trained from scratch on the RTX 3060.** The first, TxGPT, is a small GPT that reads an address's history as a sequence of transaction tokens. It scores how surprising each new transaction is, and it is the main HD model. The second is a 152M-parameter text model, pretrained on 2B tokens of web text and then taught to chat and to explain flags. It becomes the Explorer Assistant, which Assignment 3 can use.
6. **A prediction interface.** `predict.py` plus a command-line tool, so every model can score new input. The README shows how, and the Assignment 3 FastAPI back end can call the same functions.

How this keeps the Assignment 1 promise. Assignment 1 ruled out claiming fraud accuracy on data with no labels (risk 13). That rule still holds. Precision and recall are reported only in two places. One is datasets whose authors supplied labels. The other is transactions we planted ourselves, where the true answer is known by construction. The Explorer's wording never changes: "*Unusual activity requires investigation; it does not indicate wrongdoing.*"

The dates that matter.

Date	Milestone
Wed 21 Oct	Change request CR-06 sent to the team and the tutor. Snapshot DS1 fetched. Text pretraining started on the GPU.
Sun 25 Oct	M1: baseline, Isolation Forest and LOF score every snapshot transaction. This is Assignment 1 Objective 2, due 26 Oct.
Thu 29 Oct	M2: every model trained and evaluated; results frozen (version 1).
Sat 31 Oct	M3: code feature-complete; notebooks re-run from clean.
Wed 4 Nov	M4: a teammate rebuilds everything from the README on a clean machine; code frozen.
Sat 7 Nov	M5: everything submitted on Canvas. Sunday 8 Nov is spare.

The critical path. DS1 fetch → features → time split and baseline → models → evaluation → method selection → report evaluation section → submission. Both language models run on a parallel GPU lane with about three days of slack. If the text model's pretraining runs long, a decision gate cuts its token budget (Section 5.3). It never delays the critical path.

2 What is submitted and how it is marked

2.1 The files that go to Canvas

One member submits everything on behalf of the group. The brief wants files named after the group, so this plan keeps the Assignment 1 pattern.

File	Contents
`HCMC_COS30049.1_4_Assignment2_Report.pdf`	The report: at most 3,000 words, not counting the cover, contents, bibliography or appendices.
`HCMC_COS30049.1_4_Assignment2_Code.zip`	The processed datasets used by the final models, the README (as `.md` and `.pdf`), all source code, and the final trained models (Section 12.4).
`HCMC_COS30049.1_4_Assignment2_Meeting_Minutes.pdf`	Minutes for every week since the team formed (Weeks 1–8). The team prepares these separately.
`HCMC_COS30049.1_4_Assignment2_Contribution_Form.pdf`	Signed by all four members. The team prepares this separately.
`HCMC_COS30049.1_4_Assignment2_AI_Declaration.pdf`	The brief bans generative AI, but the tutor allowed it on condition that a declaration is included. This file is that declaration.

2.2 Where each point is earned

The table pairs each rubric line's top-band wording, quoted from the brief, with what this plan does to reach it. Section numbers point to where the work is described.

Criterion (pts)	Top band says	How this plan earns it	Evidence
Report 1: data collection and processing (5)	"Comprehensive and clear"	Covers all five sources: where each came from, its licence, how it was collected, and the challenges with their fixes. Processing covers cleaning, derived amounts, time-safe features, joins, scaling, tokenisation and checks. One pipeline figure ties it together. About 850 words, the largest share of the budget, because this line carries 5 of the 12 points.	Report §3–4; §6, §7
Report 2: data analysis (2)	"Detailed and clear"	Five findings, each written as question → plot → statistic → the decision it led to. This answers the brief's "inform the selection of the machine learning model".	Report §5; §8
Report 3: model selection (2)	"Thorough justification"	Written selection criteria, a table of the algorithms considered with reasons for keeping or rejecting each, and a rationale per model type.	Report §6; §9
Report 4: writing and structure (1)	"Exceptionally well-written, clear structure, no errors"	The brief's section order is followed exactly. Short sentences, consistent terms, and a proofread by a teammate.	§13
Report 5: referencing (1)	"All sources correctly cited in Harvard style"	A numbered Harvard list in the Assignment 1 style. Every dataset, library and method is cited, and a script checks that every citation is listed and every listed source is cited.	Appendix C
Report 6: completeness (1)	"All required sections included"	A checklist of the brief's sections: title, group identities including the tutor's name, introduction, problem framing, data collection, data processing, model selection, technical implementation, implementation evaluation, conclusion, bibliography and appendix.	§13
Code 1: core functionality (10)	"Data thoroughly processed; models well-selected and evaluated"	One command rebuilds features, trains every model and writes every evaluation table. Unit tests prove the time-safety rules. Evaluation results are stored as files, not only printed.	§7–§11

Criterion (pts)	Top band says	How this plan earns it	Evidence
Code 2: datasets (4)	"Complete, highly relevant, clear format"	Only the data the final models use. A data dictionary covers every file and column, columns carry units in their names, a manifest records SHA-256 checksums, and licences are recorded.	§12.4
Code 3: README (2)	"Comprehensive instructions for env setup, training, and prediction"	Conda commands, a quick check, a one-command rebuild with expected outputs and timings, prediction examples for every model, and troubleshooting.	§12.3
Code 4: comments (2)	"Thoroughly commented"	Every module, class and function has a docstring. Comments explain *why* (time-safety, leakage, thresholds), not just what.	§12.2
Code 5: structure (2)	"Clean, well-organized, modular"	One Python package with a module per pipeline step, a CLI, one configuration file and a test folder. Notebooks only call the package.	§12.2
HD: additional datasets (4)	"Utilize large number of additional datasets"	Five sources beyond the core snapshot: DS2, our own large crawl; DS3 Elliptic; DS4 BitcoinHeist; DS5 FineWeb-Edu and DS6 smol-smoltalk for the text model. DS7 is an instruction set derived from our own pipeline. Each one is used by a final model.	§6
HD: additional ML implementations (4)	"Appropriate methods, well-tuned, excellent performance"	Beyond the required types: Isolation Forest, LOF, the baseline, gradient boosting, TxGPT trained from scratch, and the 152M text model trained from scratch. Each is tuned with a recorded search and compared with a simple baseline.	§9, §10

2.3 Hard constraints from the brief

- **Word limit.** 3,000 words, excluding the cover page, table of contents, bibliography and appendix. The plan aims for 2,850 so the count never becomes an argument.
- **Group identities.** The group name and ID, every student's name and ID, *and the tutor's name*. The Assignment 1 cover had no tutor name. Action: confirm the name with the team by 22 October.
- **Python.** pandas, scikit-learn and matplotlib, as the brief names. PyTorch is added only for the two language models.
- **README.** It must explain the set-up *using conda commands*, then data processing, training and prediction.
- **Datasets in the zip.** "Must only include the data used in your final version machine model." Raw downloads that no final model uses stay out, with a script and checksum that recreate

them.

- **Late penalty.** 10% of the available mark per calendar day, and zero after seven days. The plan submits on 7 November.
- **Harvard style**, with every external statement acknowledged.
- **Generative AI.** The brief bans it; the tutor's condition is a separate declaration. The declaration lists what AI helped with (for example code drafting and proofreading) and states that the team checked and owns everything submitted.

2.4 Promises Assignment 1 made about Assignment 2

Markers can compare the two reports. Each promise below either gets delivered or gets an explicit, recorded change.

A1 promise (where)	How A2 delivers it	Plan section
Objective 1: a snapshot of 20–30 addresses (each ≥50 transactions over ≥90 days) plus 2–3 short-history addresses; a manifest; a re-runnable script (§1.1)	25 + 3 addresses using the selection rule in `config/settings.json` (seed 30049). The manifest records every DR-01 field and is reviewed by a second member.	§6.1
Objective 2: baseline, Isolation Forest and LOF score every transaction by 26 Oct 2026 (§1.1)	Milestone M1, Sunday 25 October.	§14
Objective 4: label-free trust checks in a results table, with a limitations section (§1.1, §2.3.2)	All four checks are kept: synthetic injection, stability, method agreement and the reviewer check.	§11
DR-01: source, date, selection, fields, missing values, provenance, terms, retrieval steps (§2.3)	`data/processed/esplora/manifest.json` and the data card.	§6.1, §12.4
DR-02 pass condition 1: a unit test recomputes F1–F4 for 20 random transactions from earlier records only	`tests/test_features_time_safe.py`.	§7.9
DR-02 pass condition 2: no transaction ID in both the fit and the evaluation period	`tests/test_split.py`.	§7.9
DR-02 pass condition 3: the report states how addresses were chosen and the resulting bias	Report §3 plus a limitations paragraph.	§13
DR-02 review: a code review by a member other than the author	Le Quang Anh Hoang reviews `features.py` and `split.py`.	§7.9
NFR-02: another member reproduces the pipeline from documented steps	Milestone M4: a clean install by a teammate on Windows.	§12.3

A1 promise (where)	How A2 delivers it	Plan section
The 3.5 threshold "is tuned in Assignment 2" (§2.3.1)	Threshold sweep against injection recall and the sanity rule, logged as CR-03 if it changes.	§9.3
Refinements CR-02, CR-04 and CR-05 "adopted only if approved" (§2.3.1)	An ablation compares the method as written with the refined one. The evidence goes to the team for a decision.	§7.2
LOF in novelty mode, fitted on the earlier period; flag the top 2% (§2.3.1)	Kept exactly as written.	§9.7
Sanity rule: no address with more than 20% of its transactions flagged (§2.3.2)	Checked automatically for every method.	§11
Selection rule: highest injection recall among methods meeting the stability target (§2.3.2)	Applied, and extended to the new candidate methods.	§11.4
Walkthrough results and revisions "reported with Assignment 2" (§10.4, team action T7)	A short appendix, if the UX lead ran the sessions; otherwise the report says plainly that they have not run.	§13
High-severity usability issues fixed before the A2 submission (§1.1)	Table 10.2 had none above severity 2, so this is reported as met.	§13
Risk 13: no fraud precision or recall without valid labels	Kept. Metrics on the labelled datasets are worded as "against the dataset's labels".	§4.3

3 Starting point

3.1 What already exists

Asset (location)	State and how A2 uses it
`asm1_demo/explorer/esplora.py`	A polite Esplora client using the standard library only. It pages with the last-seen-txid cursor, waits between requests, backs off on HTTP 429 and 5xx, and keeps request statistics. Reused unchanged for DS1 and DS2.
`asm1_demo/explorer/amounts.py` and `tests/test_amounts.py`	Received, sent and net per address, with 9 passing tests. The D1 check showed that summing them reproduces the API's own `funded_txo_sum` and `spent_txo_sum` (risk 18). Reused unchanged.
`asm1_demo/config/settings.json`	Every threshold in one file: selection rule, windows, minimum history, MAD scale, flag at 3.5, fit fraction 0.7, top 2%, sanity 20%, banned words. It becomes the A2 configuration and gains new sections for the models.
`asm1_demo/frontend/src/api/mock/scoring.ts`	A complete reference implementation of F1–F4 and the baseline, including CR-02, CR-04 and CR-05 and the same-block rule. The Python `features.py` is a line-by-line port, so the prototype and the pipeline agree.
`asm1_demo/scripts/d1_check.py`, `reports/d1-check.md`	One address (59 transactions) checked end to end. Its seeded block-sampling logic is the starting point for the DS1 selection script.
`asm1_demo/frontend/src/api/types.ts`	The JSON shapes the A3 front end expects. `predict.py` returns the same shapes.
`asm2/lm/` (model, tokenizer, data, train, chat, generate, smoke)	A GPT with RoPE, RMSNorm and SwiGLU, its own BPE tokenizer, memory-mapped data, masked chat fine-tuning, a speed benchmark, and Ctrl+C save and resume. It passes 15 tests and an 8-stage smoke test on CPU. It has not yet run on the GPU. The same model class serves TxGPT.
`asm2/configs/text_150m_pretrain.json`, `text_150m_chat.json`	152.6M parameters (18 layers, $d = 768$, 12 heads, sequence 1,024). 2B pretraining tokens, then a 100M-token chat fine-tune.

3.2 What is missing

1. **The real snapshot.** No folder contains the 25 + 3 address snapshot. The A1 Gantt chart planned WBS 1.2.1 for 11–13 October, so the critical path is about one week behind. The work itself is short (Section 6.1: under two hours of fetching), so this is recoverable if it starts on 21 October.
2. **The whole scikit-learn pipeline.** There are no features, models, evaluation, notebooks or figures yet.
3. **Model types that match the brief.** Isolation Forest and LOF are outlier detectors, so a literal reading of "at least two types (classification, clustering, regression)" may not credit them. Decision D-A2-1 fixes this.
4. **Labelled data.** With no labels, the brief's accuracy, precision, recall and F1 cannot be reported honestly. Decision D-A2-3 fixes this.
5. **More datasets.** A1 used one source. The brief says one simple dataset earns "a basic score".
6. **Any GPU run of the language models**, and any transaction-sequence code at all.

3.3 What the unit's own material says to do

The weekly material points in the same direction as this plan, and the report can say so:

- **Week 4 workshop.** Clean, anonymise sensitive fields, engineer security features, handle imbalance with class weights, and prefer recall-oriented metrics. Its closing self-review asks for a *data card* and a *model card*; both are in Section 12.4.

- **Week 5.** Extract, transform, load; normalise and standardise; join and concatenate; visualise class balance, distributions, correlations and time series. Its licence slide says HD students should use additional datasets and respect their licences.
- **Week 6.** Regression is evaluated with MSE, RMSE, MAE and R^2; K-Means with inertia, silhouette and Davies–Bouldin. The workshop calls outliers "not always attacks but worth investigation". It gives the workflow "cluster → sample validation → derive patterns → propose simple rules". This is the project's framing in the lecturer's own words.
- **Week 7.** DBSCAN with MinPts $= 2 \times D$ and a k-distance graph for ε, with noise points read as anomalies. Logistic regression and KNN on standardised data, scaled with the training set's parameters only. Accuracy, precision, recall, F1 and the confusion matrix. Decision trees and random forests appear as "additional" models.

4 Key decisions

Six decisions shape the rest of the plan. Each one changes what Assignment 1 approved, so all six go into change request CR-06 (Appendix A). The A1 change-control rule (§8.5) applies: all four members approve it, and because it touches the brief's marking requirements, the tutor sees it first. CR-03 is already reserved for threshold changes after the sanity check, so the next free number is CR-06.

4.1 D-A2-1: one model family for each type the brief names

Decision: cover all three types; keep the A1 methods as extras

Brief type	Models	Data	Main metrics
Clustering	K-Means (k chosen by elbow, silhouette and Davies–Bouldin); DBSCAN (ε from the k-distance graph). Distance from the nearest centre, or DBSCAN noise, marks a transaction as unusual.	DS1 (and DS2 for address profiles)	Silhouette, Davies–Bouldin, Calinski–Harabasz; recall on planted transactions
Regression	Ridge, random forest regressor, histogram gradient boosting (including quantile loss). Each predicts a transaction's amount and gap from its past; a large residual is a surprise.	DS1	MAE, RMSE, R^2 against a rolling-median baseline; coverage of the 90% interval
Classification	Logistic regression, KNN, random forest, histogram gradient boosting	DS3 Elliptic, DS4 BitcoinHeist	Accuracy, precision, recall, F1, PR-AUC, confusion matrix, all against the dataset's labels
Extra (HD)	Robust-z baseline, Isolation Forest, LOF (novelty)	DS1, plus DS3 for the label check	A1's label-free checks; PR-AUC against DS3 labels
Extra (HD)	TxGPT, a transaction-sequence GPT trained from scratch	DS2 to train, DS1 to score	Held-out perplexity against n-gram baselines; injection recall; stability
Extra (HD)	Explorer Assistant, a 152M text model trained from scratch	DS5, DS6, DS7	Validation loss; faithfulness of its explanations

Why this mapping. Clustering and regression fit the real problem: unlabelled histories in which "unusual" means unlike this address's own past. Classification needs labels, and only DS3 and DS4 have them, so that is where it runs. Doing all three types is cheap with scikit-learn. It also removes the risk of a marker deciding that two outlier detectors are "two of the same type" without the strong justification the brief demands.

Rejected alternative. Training a classifier on DS1 to recognise our own planted transactions. Precision and recall would be easy to compute, but the classifier would only relearn the injection generator. That is circular, and a careful marker would notice. It stays in the report's "considered and rejected" table.

4.2 D-A2-2: datasets

ID	Dataset	Size and labels	Licence	Used by; in the zip?
DS1	Esplora core snapshot (our collection)	25 main + 3 short-history addresses; about 5,000 transactions[est.]; no labels	Public API; terms not published, so the retrieval date and rate are recorded	Baseline, clustering, regression, IF, LOF, TxGPT scoring, evaluation. Yes: processed CSV plus gzipped raw pages
DS2	Esplora sequence corpus (our collection)	About 5,000 address histories with ≥20 transactions each; about 1M transactions[est.]; no labels	As DS1	Trains TxGPT; address-profile clustering. Yes: compact Parquet with hashed address IDs
DS3	Elliptic Bitcoin dataset (Weber et al. 2019)	203,769 transactions, 234,355 edges, 166 anonymised features, 49 time steps; 4,545 illicit, 42,019 licit, the rest unknown	Read from the Kaggle page at download and recorded	Classification; the unsupervised label check. In the zip only if the licence allows; otherwise a download script
DS4	BitcoinHeist (UCI dataset 526; Akcora, Li, Gel and Kantarcioglu 2020)	2,916,697 address-day rows, 2009–2018; six graph features; label is a ransomware family or "white"	CC BY 4.0 (UCI page, checked 20 Oct)	Classification; leave-one-family-out test. Yes: the processed sample the final model uses
DS5	FineWeb-Edu (Penedo et al. 2024)	2B tokens taken from the published shards	ODC-By 1.0 (dataset card, checked 20 Oct)	Text pretraining. No: script, manifest and checksums only
DS6	smol-smoltalk (Ben Allal et al. 2025)	484,570 multi-turn conversations; 100M tokens used	Apache 2.0 (dataset card, checked 20 Oct)	Chat fine-tune. No: script and manifest
DS7	Explorer explanation set (derived)	About 10,000 instruction–answer pairs built from DS1 and DS2 pipeline outputs, plus about 300 glossary questions written by the team	Ours	Explorer Assistant fine-tune and evaluation. Yes
DS8	Elliptic++ actors (Elmougy and Liu 2023), *optional*	Address-level features (56) and labels	None stated on the repository; ask before redistributing	Optional address-level classification, only if time allows

ID	Dataset	Size and labels	Licence	Used by; in the zip?
DS9	Ransomwhere (Cable 2024), *optional*	Crowdsourced ransomware payment addresses with their transactions and family; one 5.6 MB JSON file	CC BY 4.0	Optional label check in our own feature space: fetch these addresses' histories with the DS1 pipeline and score them (Appendix B)

Other datasets checked. Eighteen more candidates were checked for this plan, including Elliptic2, BABD-13, ORBITAAL and three synthetic anti-money-laundering sets. Appendix B lists each one with the reason it was left out. The report's data section can use that table to show that the sources were chosen, not just found.

Wording for labelled sets. BitcoinHeist's "white" means "not known to be ransomware", not "verified clean". Elliptic's "unknown" class is the majority. The report says "labelled illicit by the dataset's authors". It never claims that an address in our snapshot is illicit.

4.3 D-A2-3: where accuracy, precision, recall and F1 are allowed

Decision: label-based metrics only where labels exist

1. **DS3 and DS4.** Full classification metrics against the authors' labels.
2. **Planted transactions in DS1.** Recall, plus precision as a *lower bound*. The planted transactions are known positives. Real transactions that a method flags count as false positives even if they really are unusual, which is why the precision is only a bound.
3. **Real DS1 transactions.** No precision, recall or accuracy, ever. Only the label-free checks and the reviewer judgement, reported as judgement.

Risk 13's trigger is reworded to "any draft claims precision, recall, F1 or accuracy for real snapshot transactions".

4.4 D-A2-4: two language models trained from scratch

The tutor said a locally trained language model is needed for HD, and that it must not be a fine-tune of someone else's model. Two models meet that:

- **TxGPT** is the one that belongs to this project. It treats an address's history as a "sentence" whose "words" are transaction tokens (direction, amount bucket, gap bucket, input and output counts, fee bucket). Its per-transaction surprisal, the negative log-probability of the transaction given everything before it, is a learned unusualness score. It is time-safe by construction because the model is causal. It is also explainable: the surprisal splits by field, so the panel can say "the amount was the surprising part".
- **The Explorer Assistant** is the one that matches the everyday meaning of "LLM": a 152M-parameter text model. It is pretrained on DS5, taught to chat on DS6, and then taught on DS7 to turn a flagged transaction's features into a plain-language explanation that includes the limitation line. A3 can show it, but only behind a validator that checks every number against the input and falls back to the template text if any number is wrong.

Needs a teammate

Ask the tutor, in the CR-06 message, whether TxGPT on its own satisfies "train an LLM", or whether a text model is expected. The plan builds both, but if the answer is "TxGPT is enough", the text model drops from *Must for HD* to *Should*, and its GPU time protects the schedule instead.

Update, 22 October: a fine-tuned 3B model is added. The tutor, Arthur Hoang, agreed that the team may also fine-tune a pretrained base model of about 3B parameters. It is *added* alongside the two from-scratch models and replaces neither: the 152M Assistant finishes all three stages, and TxGPT is unchanged. The 3B model is QLoRA-tuned on the same chat and DS7 data after the from-scratch work (Section 10.3), and the report compares the two Assistants on the same measures. The tutor's agreement is confirmed, and the team approved this addition as an amendment to CR-06. The base model was first Llama 3.2 3B; on 23 October it changed to Qwen2.5-3B-Instruct, which is not gated and needs no access token (Qwen Research License: non-commercial use, a NOTICE line and "Built with Qwen"). The team accepted the change (it delegated the open A2 decisions to Cao Tri Tran on 23 October).

4.5 D-A2-5: one repository and two environments

- `asm2/` becomes the single A2 repository. The `explorer` package and `settings.json` are copied in from the prototype repository, keeping their history in the commit message. It is initialised as a git repository on 21 October and pushed to the team's GitHub so that teammates can review it. The zip is built from a tagged commit.
- `environment.yml` is **CPU-only**, the one markers use. It holds Python 3.11, pandas, NumPy, scikit-learn, matplotlib, seaborn, PyArrow, joblib, Jupyter, pytest, `tokenizers`, and CPU PyTorch for scoring with the shipped TxGPT weights. Every result can be rebuilt with it, without a GPU.
- `environment-gpu.yml` is today's file, renamed. It adds CUDA 12.6 PyTorch and is only needed to *retrain* the language models.
- `requirements.txt` is the pip fallback that A1 risk 9 promised.

4.6 D-A2-6: change control

CR-06 (Appendix A) records D-A2-1 to D-A2-5 together with their effect on the WBS, the schedule and the risks. Approval is the four members' agreement in Discord, recorded in the next minutes, with a message to the tutor before work depends on it. CR-02, CR-04 and CR-05 (the baseline refinements) are decided separately, after the ablation in Section 7.2, on evidence rather than in advance.

5 The machine and the compute budget

5.1 What the hardware can do

Part	Spec	What it means for this plan
CPU	Intel Core i5-12600K: 6 performance + 4 efficiency cores, 16 threads	scikit-learn work runs here with `n_jobs=12`, leaving threads free to feed the GPU. Elliptic tuning takes hours, not days.
GPU	NVIDIA RTX 3060, 12 GB GDDR6, Ampere, bf16 supported	Enough memory to pretrain the 152M model with micro-batches of about 8 × 1,024 tokens, no gradient checkpointing needed[est.]. TxGPT is small enough to train in minutes.
RAM	16 GB DDR4-3600	Token files are memory-mapped, never loaded whole. Heavy CPU jobs (tokenising, random-forest tuning) are scheduled so they don't overlap.
Disk	Unknown; **needs at least 40 GB free**	FineWeb shards (2.15 GB each, deleted after use), 4 GB of tokens, about 2 GB per resumable checkpoint, plus the datasets.
OS	Windows	`torch.compile` stays off (`"compile": false`) because Triton isn't supported natively on Windows. That costs roughly 20–30% speed[est.] and avoids a fragile set-up.

5.2 How long the text model's pretraining takes

For a decoder-only transformer, each training token costs about $6N$ floating-point operations for the weights plus $12\,L\,d\,T$ for attention, where N is the parameter count, L the number of layers, d the model width and T the sequence length:

$$6(152.6 \times 10^6) + 12 \cdot 18 \cdot 768 \cdot 1024 \approx 9.2 \times 10^8 + 1.7 \times 10^8 \approx 1.09 \times 10^9 \text{ FLOPs per token.}$$

For 2B tokens that totals 2.2×10^{18} FLOPs. Our estimate of the RTX 3060's dense bf16 throughput with FP32 accumulation is about 25 TFLOPS peak. A small model in eager mode typically reaches 30–40% of peak, which is 7.5–10 TFLOPS. The run therefore takes **about 60–80 hours (2.5–3.3 days)**[est.]. This matches the README's estimate of 2.5–4 days.

5.3 Decision gate on pretraining length

The benchmark (`python -m lm.train -config configs/text_150m_pretrain.json -bench`) replaces that estimate with a measured tokens-per-second figure on the first evening. Then:

Decision: fit pretraining into 72 hours of GPU time

- If the measured speed projects 2B tokens in 72 hours or less, keep 2B tokens.
- If not, set `total_tokens` to (60 hours × measured tokens per second). For example, 7,000 tokens/s gives about 1.5B tokens. The learning-rate schedule follows the new total automatically. The report states the token count actually used.
- Floor: 1B tokens. Below that, the text model isn't worth its GPU time, and the plan falls back to TxGPT as the only language model. That fallback is also recorded in CR-06.

5.4 Compute budget per job

Job	Device	Time[est.]	VRAM	RAM	Disk
DS1 fetch (28 addresses, at most 40 pages each)	Network	0.5–1.5 h	–	<1 GB	300 MB raw, 30 MB gzipped

Job	Device	Time[est.]	VRAM	RAM	Disk
DS2 crawl (about 5,000 histories)	Network	8–12 h, overnight	–	<1 GB	60 MB compact rows
DS3 and DS4 downloads	Network	<30 min	–	–	1 GB
FineWeb-Edu preparation (download, tokenizer, 2B tokens)	CPU + network	1.5–3 h	–	4–6 GB	4 GB tokens
Text pretraining	GPU	60–80 h (gated at 72 h)	7–10 GB	3–5 GB	4 GB checkpoints
Chat data preparation	CPU	10–20 min	–	2–3 GB	2 GB
Chat fine-tune (100M tokens)	GPU	3–4 h	7–10 GB	3 GB	2 GB
DS7 fine-tune and evaluation	GPU	<1 h	7 GB	2 GB	–
TxGPT sweep (3 sizes × 3 seeds)	GPU	1–3 h	1–3 GB	2 GB	<200 MB
TxGPT scoring of DS1	GPU or CPU	minutes	<1 GB	<1 GB	–
DS1 features, baseline, clustering, regression, IF, LOF	CPU	<30 min per full run	–	<2 GB	–
Synthetic injection (5 seeds × 150 plants × all methods)	CPU	30–90 min	–	<2 GB	–
Elliptic tuning (4 models, time-aware search)	CPU, 12 threads	2–6 h, overnight	–	4–6 GB	–
BitcoinHeist models (3 split schemes)	CPU	1–3 h	–	3–6 GB	–

5.5 Running four lanes at once

The machine's parts work in parallel, and the schedule (Section 14) is built around that:

- **GPU lane.** Text pretraining runs from the evening of 21 October to about 25 October. It is paused with Ctrl+C, which saves a checkpoint, whenever TxGPT needs the GPU, and resumed afterwards. Resuming is already tested. Then come the chat fine-tune and the DS7 fine-tune.
- **Network lane.** The DS1 fetch, then the DS2 overnight crawl and the DS3 and DS4 downloads. These are I/O-bound and barely touch the CPU.
- **CPU lane.** Features, every scikit-learn model, injections and evaluation. Feeding the GPU needs only one or two threads, so these run alongside pretraining.
- **Writing lane.** Notebooks, figures, the README and the report.

5.6 Windows housekeeping before a multi-day run

1. Power plan set to High performance. Sleep and hibernate off. Windows Update paused for two weeks, so a forced restart can't kill the run. Checkpoints every 30 minutes limit any loss.
2. `nvidia-smi` confirms the driver and CUDA. Watch the GPU temperature once an hour on

day 1; below about 83 °C is fine for this card.

3. `set PYTHONUTF8=1` in the Anaconda Prompt, so file encodings behave.
4. `train.py` already logs to `runs/<name>/log.jsonl`: one line per step (train loss, learning rate, gradient norm) and one per evaluation (validation loss). The loss-curve figures read this file, so it is kept with each run.
5. Nothing else uses the GPU during pretraining. Games and video editing would halve the speed or cause out-of-memory errors.

6 Workstream 1: data collection

Report criterion 1 is worth 5 points, the most of any report line, and it covers both collection and processing. Every step below therefore records what the report needs: the source, the method, the date, the numbers, and the problems hit along the way.

6.1 DS1: the core snapshot

Purpose. The 25 + 3 addresses that the Explorer shows, and the data every method is evaluated on. This is A1 Objective 1.

Selection rule. It is written into the manifest *before* any history is fetched, which avoids A1 risk 19, choosing addresses because they already look odd. All parameters come from `settings.json`:

1. Pick 20 block heights spread evenly between tip − 4,000 and tip − 100. That covers about four weeks of blocks and avoids reorganisations. The seed is 30049.
2. From each block, take the addresses on the outputs of up to 4 pages of transactions (25 per page) to form the candidate pool. Duplicates are removed and script types recorded.
3. Shuffle the pool with seed 30049 and visit candidates in that order. For each, call `/address/<a>` once (cheap) and read `chain_stats.tx_count`.
4. **Main addresses:** keep a candidate if it has 50–1,000 confirmed transactions and its full history spans at least 90 days (checked after fetching). Stop at 25.
5. **Short-history addresses:** keep the first 3 candidates with 3–9 transactions. These exercise the insufficient-history state.
6. Every visited candidate is logged with the reason it was kept or skipped (too few, too many, span too short, fetch error). The skip counts go into the report as evidence that the rule was applied.

Fetching. `python -m explorer.cli snapshot` pages each history with `/address/<a>/txs/chain[/<last_txid>]`, newest first, 25 per page, until a page holds fewer than 25 transactions or the cap of 1,000 is reached. It waits at least 0.5 s between requests and backs off exponentially on 429 and 5xx, honouring `Retry-After`. Each raw page is saved unchanged as `data/raw/esplora/<address>/page-NNN.json.gz`, so a re-run reads the cache instead of the network unless `-refetch` is given.

Verification, run automatically after the fetch and printed into the manifest:

- Number fetched equals `chain_stats.tx_count`.
- No duplicate txids, every transaction confirmed, and every block time present.
- Summed `received` equals `funded_txo_sum`, and summed `sent` equals `spent_txo_sum`, for every address (risk 18).
- For 2 addresses, the first page from the fallback (mempool.space) matches in time and amount.

Needs a teammate

A1 promised that a second member would verify one transaction's amount and direction by hand against the raw inputs and outputs, reproduce the retrieval, and review the manifest. Ask Cam Dan Bui (WBS 1.2.1 co-owner) to do this on 22 October and record their name and the date in the manifest.

Manifest (`data/processed/esplora/manifest.json`), with every DR-01 field: source and API base, retrieval timestamps, the request pace and the request statistics (count, 429s, retries, median latency), the tip height, the selection rule and seed, kept and skipped candidates with reasons, the fields used, missing-value counts, provenance (raw page paths and SHA-256), the terms note, and rebuild instructions.

Expected output[est.]. 28 addresses and about 5,000 transactions (median about 150 per address), with spans of 3–24 months.

6.2 DS2: the sequence corpus for TxGPT

Purpose. A language model needs far more sequences than 25 addresses provide. DS2 is our own large crawl, and it counts as an additional dataset.

Discovery. The same block-sampling method as DS1, over 200 blocks spread across the last 12 months, which gives seasonal variety. That yields a pool of around 60,000 distinct addresses[est.]. DS1's 28 addresses are removed from the pool before anything else happens, so TxGPT never trains on the addresses it later scores (Section 7.8).

Inclusion. 20–1,000 confirmed transactions. The target is 5,000 addresses, the minimum 3,000, the stretch 10,000.

Compact storage. DS2 does not keep raw pages. Each page is reduced at once to one row per transaction, which keeps DS2 near 60 MB instead of tens of gigabytes:

Column	Type	Meaning
`addr_id`	string	SHA-256 of the address with a project salt, first 16 hex characters. The salt is kept out of the zip (A1 risk 16).
`txid_hash`	string	Hash of the txid, used only to remove duplicates.
`block_height`, `block_time_utc`	int	Confirmation height and Unix time.
`received_sats`, `sent_sats`	int	As `amounts.py` computes them.
`n_inputs`, `n_outputs`	int	Transaction shape.
`fee_sats`, `vsize_vb`	int	For the fee-rate token.

Operations. A state file records each address as done, partial (with its last txid) or failed, so the crawl resumes after any interruption. The pace is 0.5 s with back-off. If Blockstream returns repeated 429s, the client switches to mempool.space for an hour and then returns. Expected load[est.]: about 5,000 addresses × 4 pages on average gives 20,000 requests at about 1.2 s each, roughly 7 hours, run overnight on 22 October.

Checks. Every address's row count matches its `tx_count`, up to the cap. There are no duplicate `txid_hash` values within an address, and no DS1 address appears.

6.3 DS3: Elliptic

- **Source.** Kaggle, "Elliptic Data Set", released with Weber et al. (2019). Three CSVs: features (txId, time step, 165 features), classes (`1` = illicit, `2` = licit, `unknown`) and the edge list.

- **Download.** `python -m explorer.cli external -elliptic`, which uses the Kaggle CLI if a token is set and otherwise prints the manual steps. It records SHA-256 of each file.
- **Licence.** Read from the Kaggle page on the day of download and copied into the data card. If redistribution isn't allowed, the zip carries the download script and checksums instead of the data, and the README says so.
- **Challenges for the report.** The features are anonymised, so they can't be computed for our own addresses. That is why DS3 is used to evaluate method families, not to score DS1. About 77% of nodes are unlabelled. The authors report that a dark-market shutdown around time step 43 changes the data distribution.

6.4 DS4: BitcoinHeist

- **Source.** UCI Machine Learning Repository, dataset 526, from Akcora et al. (2020). A1 already cites the UCI page. Licence CC BY 4.0 (checked 20 October 2026); a 110.7 MB zip holding one 225 MB CSV: `address, year, day, length, weight, count, looped, neighbors, income, label`.
- **Download.** `python -m explorer.cli external -bitcoinheist`, with SHA-256 recorded.
- **Challenges for the report.** Severe imbalance (positives are about 1.4% of rows[est.]). Labels come from three earlier ransomware studies, so "white" isn't verified innocence. Features are 24-hour graph statistics, not single transactions, which is why A1 called it a comparison source. Family coverage differs by year, which the time-split design must respect.

6.5 DS5 and DS6: text corpora

Collected by the existing scripts. `python -m lm.prepare_text -delete-raw` streams FineWeb-Edu shards, trains our 32,768-token BPE tokenizer on the first 100,000 documents, writes 2B training tokens plus 10M validation tokens, and writes `meta.json`. `python -m lm.prepare_chat` renders smol-smoltalk with our chat template. Conversations that are too long are truncated at an assistant turn rather than dropped (about 46% exceed 1,024 tokens). It writes the token and loss-mask files. Both scripts already record source, revision, counts and settings, which the data card reuses.

6.6 DS7: the explanation set

Built on 29 October, after the models exist, by `python -m explorer.cli build-ds7`:

- **About 10,000 explanation pairs.** The input is a JSON summary of a scored transaction from DS1 or DS2: direction, amount, the usual amount and gap, each feature's z, the comparison basis, and which features contribute. The output is the Explorer's template explanation, the same wording rule as the A1 detail panel, ending with the fixed limitation line. Five phrasings of each template stop the model memorising one sentence.
- **About 300 glossary pairs written by the team**, for example "What does a robust z of 4 mean?" or "Why isn't an unusual transaction proof of wrongdoing?". These are written from our own report text.
- **Split by source address**: 90% train, 5% validation, 5% test, so test pairs never come from a training address.

6.7 Collection problems and responses, for the report

Problem	Response
Esplora pages hold 25 transactions and rate limits aren't published	Cursor paging; a pause of at least 0.5 s; exponential back-off honouring `Retry-After`; raw-page caching; request statistics logged in the manifest.
Raw transaction JSON is large (about 1.3 MB per page for busy addresses)	DS1 keeps raw pages gzipped as evidence; DS2 reduces each page to compact rows immediately.
Very busy addresses (exchanges) exceed the 1,000-transaction cap	Excluded by the rule; stated as a bias (the sample leans towards ordinary wallets).
Address-level amount and direction aren't API fields	Derived from `vout` and `vin.prevout`; verified by the sum checks and a manual check.
Selection bias	Rule fixed before fetching; seeded; all skips logged.
No ground-truth labels for Bitcoin addresses	Two labelled public datasets (DS3, DS4) for label-based metrics; planted transactions for known positives.
Labelled datasets are anonymised or aggregated	Used to evaluate method families, not to score DS1.
Licences and privacy	Licences recorded per dataset; DS2 addresses hashed; no identity claims.

Done when

DS1 holds 25 + 3 addresses and passes every automated check, with the manifest reviewed by a teammate. DS2 holds at least 3,000 addresses and passes its checks. DS3 and DS4 are downloaded with checksums and licences recorded. DS5 and DS6 are tokenised. Collection numbers (requests, 429s, skips) are copied into `reports/tables/collection_summary.csv`, ready for the report.

7 Workstream 2: processing and feature engineering

7.1 DS1: from raw pages to a clean transaction table

`python -m explorer.cli transactions` turns the raw pages into `data/processed/esplora/transactions.csv`, one row per (address, transaction). Each step logs how many rows it removed, and that log becomes the processing table in the report:

1. Parse every page and keep confirmed transactions only. The chain endpoint returns only confirmed ones, but the check is explicit anyway.
2. Remove duplicate (address, txid) pairs, which can occur at page boundaries if a new block arrives mid-fetch.
3. Derive `received_sats`, `sent_sats`, `net_sats` and `direction` with `amounts.address_flow`. Also derive `n_inputs`, `n_outputs`, `fee_sats`, `vsize_vb` and `block_time_utc`.
4. Drop transactions where the address neither receives nor sends (expected count: 0). A non-zero count is logged as a data problem.
5. Sort by (address, block time, txid). Transactions in the same block share a time and are never part of each other's history.
6. Units are always in the column names (`_sats`, `_utc`, `_hours`). Amounts stay integers in satoshis; BTC appears only in plots and in the interface.

Missing values, recorded per column. The gap (F3) doesn't exist for an address's first transaction, so it stays empty rather than being filled with an invented number, and the transaction isn't scored. Fees are always present for confirmed transactions. Any other missing value stops the pipeline with a clear message, because it would mean the API changed.

7.2 Features F1–F4: a port of the prototype

`explorer/features.py` is a line-by-line Python port of `scoring.ts` and reads the same `settings.json`. That keeps A1 §2.3, the prototype and the pipeline consistent.

ID	Feature	Definition (history = strictly earlier transactions of the same address)	Scale for the z-score
F1	Size	$\log_{10}(1 + \lvert\text{net}\rvert)$, in satoshis	as is
F2	Burst	number of earlier transactions in the previous 7 days	as is (count)
F3	Gap	hours since the previous earlier transaction	$\log_{10}(1 + h)$
F4	Relative size	$\lvert\text{net}\rvert \div$ median $\lvert\text{net}\rvert$ of the size window	$\log_{10}(\max(r, 0.001))$

Comparison windows (CR-02): the previous 30 days, extended back until at least 20 earlier transactions are included. The *timing window* holds all earlier transactions and is used for F2 and F3. The *size window* holds only earlier transactions in the same direction (CR-04) and is used for F1 and F4.

Robust z for each feature, against the same feature over its window, with a floor on the spread (CR-05):

$$z = \frac{x - \text{median}(W)}{\max\big(1.4826 \cdot \text{MAD}(W),\ s_{\min}\big)}, \qquad s_{\min} = (0.1,\ 1,\ 0.1,\ 0.1) \text{ for F1–F4.}$$

Scorable means at least 10 earlier transactions spanning at least 14 days, and at least 5 values in the window. Unscorable transactions are kept and displayed, but never flagged.

Extended features (used only in an ablation and as TxGPT tokens, never in the Explorer's explanation): F5 $\log_2 n_{\text{inputs}}$, F6 $\log_2 n_{\text{outputs}}$, F7 $\log_{10}$ fee rate (sat/vB), and F8 the hour of day as (sin, cos).

Inputs to pooled models. One model is fitted across all 25 addresses, which hold very different amounts: a tip jar next to a merchant. So the clustering, IF and LOF inputs are each transaction's *own-history z-scores* $(z_1, \ldots, z_4)$, clipped to ± 20 so that a single extreme value can't dominate, plus a one-hot direction. This makes "unusual for this address" comparable across addresses. The ablation repeats the models on transformed raw features scaled with a `StandardScaler` fitted on the fit period.

Refinement ablation (CR-02, CR-04, CR-05). The baseline and every downstream model run twice: once with the method as A1 §2.3 wrote it (a 30-day window, all directions, no transform, no floor), and once with the refinements. For each run the evidence recorded is the injection recall per pattern, the share of addresses that break the 20% sanity rule, and the flag rate. The team adopts the refinements in the Change Log only if the evidence supports them. Either way, the report states which version is final and why.

7.3 The time split

For each address, take the scorable transactions in time order. The cut-off time t^* is the time of the transaction at the 70th percentile (`fitFraction` $= 0.7$). Transactions with time $< t^*$ form the **fit period**; those with time $\geq t^*$ form the **evaluation period**. Same-block transactions therefore never straddle the cut. Unscorable early transactions serve only as history. Every model is fitted on fit-period rows pooled across addresses. Every threshold, such as the top-2% cut, is set from fit-period scores and then applied unchanged to the evaluation period.

7.4 Synthetic injection: known positives in real histories

`explorer/inject.py` plants patterns in a *copy* of one address's history, inside its evaluation period. Features are then recomputed time-safely, because a planted transaction changes the features of

everything after it. Finally the copy is scored with *frozen* models. A single plant goes into each copy, so plants never mask one another.

Pattern (A1 §2.3.2)	**How it is planted**
Value spike	One transaction at a random time between two real ones. Its direction is the address's majority direction over the previous 30 days. Its amount is 10 × that direction's median |net| up to that point. Input and output counts are the address's median.
Burst	Ten transactions within a random 24-hour period, with amounts drawn from the address's own earlier amounts, so only the timing is unusual. Scored two ways: per transaction (A1's measure, the share of the 10 flagged) and per event (at least one flagged).
Dormant wake-up	One transaction appended 60–120 days (uniform) after the address's last real one, with a typical amount, so only the gap is unusual.

There are 50 plants per pattern, spread across addresses in proportion to their evaluation-period size, and the whole set is repeated with 5 seeds (30049 to 30053). Results are reported as mean ± standard deviation. Each planted row carries `is_synthetic = 1` and can never leak into a fit set.

7.5 DS3 Elliptic processing

1. Join features to classes on `txId` (a left join; every feature row has a class row). Map `1` to illicit (1), `2` to licit (0), and `unknown` to missing.
2. Name the columns `time_step`, `local_01`–`local_93` and `agg_01`–`agg_72`.
3. **Join the edge list** to add two engineered graph features, in-degree and out-degree per transaction. This is a genuine dataset join, one of the processing steps the brief lists.
4. Feature sets: **LF**, the 93 local features (+2 degrees), and **AF**, all 165 (+2). `time_step` is never a feature; it only defines the split.
5. **Temporal split** as in Weber et al. (2019): time steps 1–34 for training and 35–49 for testing. Tuning uses only training time steps: fit on 1–26, validate on 27–34.
6. Scaling inside a scikit-learn `Pipeline`, so the scaler only ever sees training rows. Check for missing and constant columns and drop constants.

7.6 DS4 BitcoinHeist processing

1. Binary label: `white` → 0, any family name → 1. The family is kept for the leave-family-out test.
2. $\log_{10}$ `income` (satoshis). Apply $\log(1 + x)$ to `weight`, `count`, `looped` and `neighbors` if the analysis confirms heavy tails.
3. `address`, `year` and `day` are *not* features. The address is an identifier. Year and day would let a model learn when each family was active instead of what the activity looked like, which is a leakage shortcut. They only define splits.
4. Remove exact duplicate rows and record how many were removed.
5. **Splits**, fixed after the analysis by a written rule. (a) Temporal: train on years up to Y and test on later years, with Y the latest year that leaves at least 1,000 positives in the test set. (b) A stratified random 80/20 split as an optimistic reference. (c) Leave one family out, for each of the five largest families.
6. **Training-set sampling.** All positives plus negatives downsampled to a 10:1 ratio (seeded), *in the training set only*. Test sets keep the real class ratio, because precision depends on the base rate.

7.7 DS2: TxGPT tokenisation

Each transaction becomes 6 tokens, always in this order. Gap comes first so the model predicts the transaction's content knowing when it happened:

Field	Tokens	Buckets
GAP	18	FIRST (no earlier transaction); SAME_BLOCK; then hours since the previous transaction, with upper bounds 1/6, 1, 3, 6, 12, 24, 48, 96, 168, 336, 720, 1,440, 2,160, 4,320, 8,760, and above
DIR	3	IN, OUT, ZERO (net 0)
AMT	47	$\lfloor \log_2 \lvert \text{net} \rvert \rfloor$ clipped to 0–46: one bucket per doubling, from 1 sat to about 700,000 BTC
NIN	7	input count: 1, 2, 3–4, 5–9, 10–19, 20–49, 50 or more
NOUT	7	output count, same buckets
FEE	9	fee rate in sat/vB: below 1, then doubling bounds 2, 4, 8, 16, 32, 64, 128, and above
Special	3	PAD, BOS (start of a history), UNK
Total	94	padded to 128 for GPU-friendly shapes (`pad_vocab` in `lm/model.py`)

- A history becomes BOS followed by $6n$ tokens, oldest first.
- Training windows are 768 tokens (128 transactions). Longer histories are cut into windows with a stride of 384 tokens, aligned to transaction boundaries. BOS appears only at a true history start, so the model learns both "beginning" and "middle" contexts.
- Split *by address*: 90% train, 5% validation, 5% test. DS1 is never in DS2.
- Written by `python -m lm.prepare_tx` as `train.bin`, `val.bin` and `test.bin` (uint8 is enough for 128 tokens), plus `meta.json` with the bucket edges, so scoring uses identical buckets.

7.8 Leakage controls in one place

1. Features use strictly earlier transactions only, and same-block transactions are excluded (DR-02).
2. The fit/evaluation split is by time within each address. Models and thresholds are fitted on the fit period only.
3. Every scaler lives inside a `Pipeline` fitted on training rows.
4. Elliptic uses a temporal split, with tuning confined to training time steps. The test period is used once.
5. BitcoinHeist uses neither year nor day as a feature; downsampling happens in the training set only.
6. TxGPT and DS7 are split by address, and DS1 is excluded from DS2.
7. Plants go only into evaluation-period copies, scored by frozen models.
8. No test number is ever used to choose a hyperparameter. If one is, the report says so.

7.9 Tests

Test file	What it proves
`test_amounts.py` (existing, 9)	Received, sent, net and direction on hand-made transactions.
`test_features_time_safe.py`	**DR-02 condition 1**: for 20 random transactions, features recomputed from a history truncated to strictly earlier transactions equal the pipeline's values exactly.
`test_features_parity.py`	The Python port matches `scoring.ts` on the prototype's illustrative histories, to within 10^{-9}. A small script dumps the TypeScript results to JSON for this.
`test_split.py`	**DR-02 condition 2**: no txid is in both periods, and the latest fit time is before the earliest evaluation time for every address.
`test_injection.py`	Plants land in the evaluation period only. A spike is 10 × the median, a burst is 10 transactions within 24 hours, and a wake-up gap is at least 60 days. Every plant is marked synthetic.
`test_tx_tokens.py`	Bucket edges, token round trips, a sequence length of $1 + 6n$, and an identical vocabulary at training and scoring time.
`test_predict.py`	Prediction outputs match the A1 API shapes, the limitation line is present on every flagged item, and no banned word appears in any user-facing string.
`tests/test_lm.py` (existing, 15) and `lm.smoke`	The language-model code end to end on CPU.

All of these run with `python -m pytest -q` in under two minutes on CPU, which is also the README's quick check.

Needs a teammate

A1 requires DR-02 code review "by a member other than the author". Le Quang Anh Hoang reviews `features.py`, `split.py` and `inject.py` on 24–25 October, against the tests above, and their review comments go in the minutes.

Feeds the report and rubric

Report §4 (Data processing): the cleaning log table, the feature table, the time split, the leakage list (condensed to 3–4 sentences), the DS3 join, the DS4 sampling and the TxGPT tokenisation. Code rubric lines 1, 4 and 5, through testable, documented modules.

8 Workstream 3: data analysis

Report criterion 2 is worth 2 points, for analysis that is "detailed and clear". The brief asks for analysis that will "uncover patterns and insights" and "inform the selection of the machine learning model". So every analysis below is written as a question, then a plot, then a statistic, then the decision it informs. The Week 5 visualisation notebook uses the same "why this plot / so what" pattern.

Where it lives. `notebooks/01_eda.ipynb` calls `explorer.plots`, so every figure is also produced by `python -m explorer.cli figures` as a vector PDF in `reports/figures/`. The report includes the PDFs directly, never screenshots.

Plot rules (A1 NFR-01): a title; axis labels with units (BTC, satoshis, hours, count, date UTC); the snapshot date in a footnote for DS1 plots; one colour per method across the whole report; a colour-blind-safe palette; log axes wherever a variable spans orders of magnitude, with that stated on the axis.

8.1 DS1 and DS2 analyses

Question	Plot	Statistic	Decision it informs
How big and how long is each history?	Bar chart of transactions per address; strip of history spans	median, IQR	Whether the 70/30 split leaves enough evaluation rows per address
What does "normal" look like for one address?	Daily timeline for 3 contrasting addresses (the Explorer's own chart)	–	Motivates per-address baselines over global thresholds
How are F1–F4 distributed?	Histograms on log axes; box plots by direction	skewness, kurtosis	Heavy tails justify the log transforms, the robust z (median/MAD rather than mean/SD) and the spread floor
Do incoming and outgoing amounts differ?	Paired box plots of |net| by direction per address	Mann–Whitney U per address	Evidence for CR-04 (same-direction size windows)
Are the features redundant?	Spearman correlation heatmap F1–F8	ρ	F1 and F4 are expected to correlate; justifies keeping both (absolute vs habitual size) or dropping one in models
Is activity periodic?	Hour-of-day and weekday histograms (UTC)	χ^2 against uniform	Whether F8 (hour) is worth testing
How are gaps distributed?	Empirical CDF of gaps, log-x	quantiles	Sets the TxGPT gap-bucket edges; shows dormancy is rare
Does the feature space have dense groups and sparse points?	2-D PCA of $(z_1..z_4)$, coloured by direction	explained variance	Non-spherical density supports DBSCAN alongside K-Means
How many transactions break $\|z\| \geq 3.5$ on the raw data?	Flag-rate bar per address	share per address	An early warning for the 20% sanity rule; starts the threshold tuning
What kinds of addresses are in DS2?	Address-level feature pairs plot (transactions per day, in-share, median size, gap CV)	–	Inputs to address-profile clustering; shows DS2 covers DS1's range, so TxGPT's training data matches its scoring data
How long are the DS2 sequences, and which tokens dominate?	Sequence-length histogram; token-frequency bars per field	–	The context length (128 transactions) and the bucket design

8.2 DS3 and DS4 analyses

Question	Plot	Statistic	Decision it informs
How imbalanced is Elliptic, and how does that change over time?	Stacked bars of illicit / licit / unknown per time step	illicit share per step	Temporal split; PR-AUC and F1 rather than accuracy; class weights
Which Elliptic features separate the classes?	Top-10 features by mutual information; violin plots	mutual information; Mann–Whitney U	LF versus AF comparison; RF importance cross-check
How imbalanced is BitcoinHeist, and in which years?	Positives per year and per family	counts	Fixes the temporal split year Y and which families get a leave-out test
Are the BitcoinHeist features heavy-tailed?	Histograms on log axes by label	skewness	The $\log(1+x)$ transforms; robust scaling
Do ransomware and white rows differ?	Box plots of log income, length, weight and count by label	Mann–Whitney U with effect size (rank-biserial)	Which features carry signal; expectations for linear versus tree models

8.3 What the report's analysis section will contain

Five findings, about 80 words each, each with one figure or a two-panel figure. The others stay in the notebook and the appendix:

1. Feature distributions are heavy-tailed, which motivates the log scales and robust z.
2. Each address has its own scale, and in and out amounts differ, which motivates per-address and same-direction baselines.
3. The z-feature space has dense cores and sparse points, which motivates DBSCAN alongside K-Means.
4. The labelled data is severely imbalanced and drifts over time, which motivates temporal splits, class weights and PR-based metrics.
5. DS2 covers the behaviour seen in DS1, which justifies training TxGPT on DS2 and scoring DS1.

The wording above is a hypothesis. If the data says otherwise, the finding changes, and the change itself is reported. That is still analysis that informed a decision.

Done when

The notebook runs top to bottom from a clean kernel. Every figure has a "so what" sentence under it. The five report figures exist as PDFs in `reports/figures/`. Statistical test results are saved in `reports/tables/eda_tests.csv`.

9 Workstream 4: the scikit-learn models

9.1 Selection criteria (the report's model-selection section)

Report criterion 3, "thorough justification", is worth 2 points. The report states its criteria first, then scores each candidate against them:

ID	Criterion
C1	Fits the data that exists: supervised learning only where labels exist; unsupervised or self-supervised elsewhere.
C2	Explainable in the Explorer's detail panel (FR-04, AC-03): the reason for a flag can be stated in plain words.
C3	Scores a new transaction without refitting (novelty scoring), so it respects the time split and works in A3.
C4	Robust to heavy tails and class imbalance (Section 8).
C5	Trains and tunes on our hardware within the schedule.
C6	Taught in Weeks 6–7, or a justified step beyond what was taught.

9.2 Candidates considered

Candidate	Type	Kept	Reason
K-Means	Clustering	✓	C2, C5, C6. Centroids describe groups in feature terms; fast; the unit's reference clustering method.
DBSCAN	Clustering	✓	C1, C4. Finds arbitrary shapes; noise points are the unusual ones; no need to choose k (Week 7).
Gaussian mixture	Clustering	✗	Assumes Gaussian components, which the heavy tails break. Adds little beyond K-Means plus IF.
OPTICS	Clustering	✗	Handles varying density, but DBSCAN on z-features already copes; time. Named as future work.
Ridge regression	Regression	✓	C2, C6. A linear, readable baseline that copes with correlated inputs (L2 penalty, Week 6).
Polynomial regression	Regression	✗	Trees capture non-linearity with less manual feature design.
Random forest regressor	Regression	✓	Non-linear, robust to outliers in the inputs, gives feature importance.
Histogram gradient boosting (regression, quantile)	Regression	✓	Strong on tabular data; quantile loss gives a 90% expected range, which is easy to explain ("outside the usual range").
Logistic regression	Classification	✓	C2, C6. Interpretable coefficients and probabilities; the reference model.
KNN	Classification	✓	C6. Classifies by similarity to known cases; must be standardised (Week 7). Subsampled for DS4.
Decision tree	Classification	✗(final)	Overfits and is unstable (Week 7). Reported only as a reference row.
Random forest	Classification	✓	Strong tabular baseline. The published reference model on Elliptic (Weber et al. 2019).
Histogram gradient boosting	Classification	✓	Usually the strongest tabular learner; built into scikit-learn, so no extra dependency.

Candidate	Type	Kept	Reason
XGBoost, LightGBM	Classification	✗	Same family as histogram gradient boosting, but adds dependencies the marker must install.
SVM (RBF)	Classification	✗	Training cost grows with the square of the rows (2.9M in DS4); sensitive tuning (C5).
Graph neural network (GCN)	Classification	✗	Outside the unit and the schedule. The obvious future work for Elliptic.
Isolation Forest	Outlier	✓(extra)	The A1 plan. Fast; novelty scoring; handles mixed scales.
Local Outlier Factor	Outlier	✓(extra)	The A1 plan. Density relative to neighbours; novelty mode (`novelty=True`).
One-class SVM	Outlier	✗	Sensitive to its kernel and ν; IF and LOF already cover the outlier family.
Autoencoder	Outlier	✗	DS1 is too small; TxGPT is the deep-learning experiment.
Classifier on planted labels	Classification	✗	Circular: it would learn the injection generator (Section 4.1).
LSTM sequence model	Sequence	✗	The transformer code already exists and trains in parallel; same role as TxGPT.
Fine-tuning a public LLM (e.g. GPT-2)	Language	✗	The tutor requires training from scratch.

9.3 Baseline: robust z (A1 §2.3)

- A transaction is flagged when it is scorable and $\max_k |z_k| \geq 3.5$. Its score is $\max_k |z_k|$, which lets the baseline join the ranking checks.
- **Threshold sweep**, which A1 promised. Try 2.5, 3.0, 3.5, 4.0, 4.5 and 5.0, and record injection recall per pattern (on the *tuning* plant set, seeds 40000–40004), the flag rate, and the number of addresses breaking the 20% rule. Choose the smallest threshold that meets the sanity rule and the recall target. A move away from 3.5 is logged as CR-03, which A1 already reserved for exactly this.
- The detail panel lists every feature with $|z| \geq 2$, as A1 specified.

9.4 Clustering: K-Means and DBSCAN

Input. $(z_1, \ldots, z_4)$ clipped to ± 20, plus the direction one-hot, so $D = 6$. Fitted on the fit period, pooled across addresses.

K-Means.

1. Try $k = 2, \ldots, 10$ with `n_init=20` and seed 30049. Record inertia, silhouette (on a 10,000-row sample if needed), Davies–Bouldin and Calinski–Harabasz.
2. Choose k by the best silhouette, checked against the elbow and Davies–Bouldin. If they disagree, the report says so and explains the choice.
3. Describe each centroid in words from its mean z values, for example "typical", "large for this address, outgoing" or "quick follow-up". These descriptions are what make the clusters useful to an analyst (Week 6 workshop, §3.4).
4. **Unusualness** is the distance to the nearest centroid. The threshold is the 98th percentile of

fit-period distances, matching A1's top 2%.

DBSCAN.

1. MinPts $= 2D = 12$ (Week 7 rule). Plot the sorted 12-nearest-neighbour distances and find the knee by the maximum distance from the chord, which takes a few lines of NumPy and no extra package. That gives ε_0.
2. Grid: $\varepsilon \in \{0.7, 0.85, 1, 1.15, 1.3\} \times \varepsilon_0$ and MinPts $\in \{8, 12, 16\}$. Keep settings whose fit-period noise share is 1–5%, and among those pick the best silhouette on non-noise points.
3. **Scoring new points.** scikit-learn's DBSCAN has no `predict`. So a new point is noise when no core sample lies within ε (a `NearestNeighbors` index on the core samples); otherwise it joins the cluster of its nearest core sample. Its ranking score is the distance to that nearest core sample. The report states this rule.

Address profiles (DS1 + DS2). One row per address with 8 features: transactions per day, in-share, median log size, gap coefficient of variation, a burstiness index, mean output count, history length and median fee rate. Standardise, run K-Means with $k = 3, \ldots, 8$ by silhouette, and describe the profiles. This is a second, address-level clustering use. It feeds the analysis section and gives A3 an optional "activity profile" label.

9.5 Regression: expected behaviour and surprise

Targets. $y_{\text{size}} = \log_{10}(1 + |\text{net}|)$ and $y_{\text{gap}} = \log_{10}(1 + \text{hours since previous})$.

Inputs, all time-safe. Direction; the rolling median and MAD of y_{size} over the same-direction size window; the rolling median and MAD of y_{gap} over the timing window; F2; the count in the previous 30 days; the previous transaction's size and direction; the address's age (log hours since its first transaction); its in-share so far; and the hour and weekday as sin/cos. When predicting size, the current amount is excluded; when predicting the gap, the current gap is excluded. A1's DR-02 allows a transaction's own time and direction.

Models.

- **Naive**: predict the rolling median. This is exactly the robust-z centre, so beating it shows what learning adds.
- **Ridge**: $\alpha \in 10^{\{-3,\ldots,3\}}$, 13 values.
- **Random forest**: 300 trees; max depth $\{\text{None}, 10, 20\}$; minimum leaf $\{1, 5, 20\}$; max features $\{1.0, \sqrt{\cdot}\}$.
- **Histogram gradient boosting**: learning rate $\{0.03, 0.1\}$; max leaf nodes $\{15, 31, 63\}$; L2 $\{0, 1\}$. Plus quantile models at 0.05 and 0.95 for a 90% expected range.

Tuning uses the last 20% of the fit period as validation (time order kept). Final models are refitted on the whole fit period.

Metrics on the evaluation period: MAE, RMSE and R^2 on the log scale; skill against the naive model, $1 - \text{MAE}/\text{MAE}_{\text{naive}}$; and the 90% interval's coverage (target about 90%) and mean width.

Unusualness. The standardised residual $r = (y - \hat{y})/s$, where s is $1.4826\times$ the MAD of that address's fit-period residuals (with a floor). The score is $\max(|r_{\text{size}}|, |r_{\text{gap}}|)$, flagged at 3.5 for consistency with the baseline. A second variant flags anything outside the 98% quantile range.

9.6 Classification on DS3 and DS4

Model	Search space	Search
Logistic regression	$C \in 10^{\{-3,\ldots,2\}}$; L2; class weight {none, balanced}; `max_iter=2000`	full grid (12)
KNN	$k \in \{3, 5, 9, 15, 25, 41\}$; weights {uniform, distance}; $p \in \{1, 2\}$	full grid (24)
Random forest	trees $\{300, 600\}$; max depth {None, 20, 40}; max features $\{\sqrt{\cdot}, 0.3\}$; min leaf $\{1, 3\}$; class weight {none, balanced_subsample}	random, 30 draws
Hist. gradient boosting	learning rate $\{0.05, 0.1\}$; max leaf nodes $\{31, 63, 127\}$; L2 $\{0, 1\}$; class weight {none, balanced}; early stopping on	random, 30 draws

Elliptic protocol.

1. A `PredefinedSplit` fits on time steps 1–26 and validates on 27–34. The selection score is F1 for the illicit class.
2. Refit the best configuration on steps 1–34. Choose the decision threshold that maximised validation F1, never the test F1.
3. Evaluate once on steps 35–49. Report accuracy, precision, recall and F1 (illicit), macro-F1, PR-AUC, ROC-AUC, the confusion matrix, and F1 per test time step. The last of these shows the drift around the dark-market shutdown.
4. Run both feature sets (LF, AF), giving 8 rows. Compare with the random-forest result reported by Weber et al. (2019); the numbers are copied from the paper and cited, not remembered.
5. Permutation importance (top 15) for the best model, computed on test data.

BitcoinHeist protocol. The same four models, the same metrics and three split schemes: temporal (validation on the last training year), stratified random (validation on 10% of training), and leave one family out (recall on the held-out family, the honest test of whether the model generalises to ransomware it has never seen). KNN trains on a 200,000-row stratified subsample, using a KD-tree in 6 dimensions.

Optional: multi-class family classification, if time allows. It is a *Could* in Section 15.3.

9.7 Isolation Forest and LOF (the A1 extras)

- Inputs and fit period as for clustering.
- **Isolation Forest**: trees $\{200, 500\}$; max samples $\{256, 1024, \text{auto}\}$; max features $\{1.0, 0.75\}$. Score = $-$`score_samples`.
- **LOF**: `novelty=True`; neighbours $\{10, 20, 35, 50\}$. Score = $-$`score_samples`.
- Flag threshold: the top 2% of fit-period scores (A1).
- **Tuning without labels.** Choose hyperparameters by recall on the *tuning* plant set (seeds 40000–40004) and by stability. The plant sets reported in the evaluation use different seeds (30049–30053), so tuning never sees the planted transactions it is scored on. The report states this.

The label check (DS3, and DS4 temporal). Fit Isolation Forest, LOF, K-Means distance and DBSCAN distance on the training period (Elliptic: all nodes including unknown, AF, scaled), then score the labelled test nodes. Report PR-AUC, precision and recall at the top 2%, and the prevalence as the chance level, with the supervised random forest as the upper reference. This

measures how far "unusual" and "labelled illicit" coincide, which is evidence for the Explorer's wording either way.

9.8 Search budget and saved artefacts

Model group	Configs	Selection signal	Where it is logged
Baseline threshold	6	tuning-plant recall + sanity rule	`reports/tables/tuning_baseline.csv`
K-Means	9	silhouette, elbow, DB	`tuning_kmeans.csv`
DBSCAN	15	noise share 1–5% + silhouette	`tuning_dbscan.csv`
Regression (3 models × 2 targets)	13 / 18 / 12 each	validation MAE	`tuning_regression.csv`
IF / LOF	12 / 4	tuning-plant recall + stability	`tuning_outliers.csv`
Elliptic (4 models × 2 feature sets)	12–30 each	validation F1 (illicit)	`tuning_elliptic.csv`
BitcoinHeist (4 models × 3 splits)	12–30 each	validation F1	`tuning_heist.csv`

Every final model is saved with `joblib` in `models/`. Next to it sits a **model card** (`<name>.card.json`) recording the training-data checksum, parameters, metrics, seed, date, and scikit-learn and Python versions. The Week 4 self-review asked for model cards.

Feeds the report and rubric

Report §6 (Model selection): the criteria table, a condensed candidates table (about 12 rows) and one paragraph per model type. Report §7.2: the tuning results go to the appendix, with only the selected settings in the body. HD "additional ML": gradient boosting, IF and LOF, the label check, and the recorded searches ("well-tuned").

10 Workstream 4b: the two language models

Both models use the existing `lm/model.py` (RoPE, RMSNorm, SwiGLU, PyTorch's fused attention) and `lm/train.py` (bf16 autocast, AdamW, cosine schedule, checkpoint every 30 minutes, Ctrl+C saves, resume on restart). Neither starts from anyone else's weights or tokenizer.

10.1 TxGPT: a transaction-sequence model

Idea. A language model learns which next "word" is likely. Here the words are the 6 tokens of each transaction (Section 7.7), so the model learns how Bitcoin addresses usually behave over time. A transaction the model finds unlikely, *given that address's own history*, is unusual. Because the model is causal, a transaction is scored only from what came before it, which is time-safe by construction.

Three sizes, to find the right one for the data. DS2 gives about 6M training tokens[est.]. That is small for a language model, so size is tuned rather than assumed:

Size	Layers	d	Heads	SwiGLU hidden	Params[est.]	Peak learning rate
S	4	128	4	352	0.8M	1×10^{-3}
M	6	256	8	688	4.7M	1×10^{-3}
L	8	384	6	1,024	14M	6×10^{-4}

Training configuration (`configs/txgpt_{s,m,l}.json`): sequence 768 tokens; batch 32 se-

quences (about 24,600 tokens); up to 20 passes over the data; validation every 100 steps; early stopping after 5 validations without improvement; warm-up 100 steps; cosine decay to 10%; weight decay 0.1; dropout 0.1, because the data is small; bf16. Each size is trained with 3 seeds, 9 runs in total, at minutes each[est.]. The size with the lowest mean validation loss wins. A plot of validation loss against parameter count goes in the appendix as evidence of tuning.

Is it learning anything? Perplexity against simple baselines. On the held-out test addresses, compare the average negative log-likelihood per token and per transaction, field by field, with three baselines that need no neural network:

1. **Global frequencies**: each field's overall token frequency.
2. **Address habit**: each field's token frequency in *this address's* history so far (Laplace-smoothed). This is close to what the robust-z baseline assumes.
3. **First-order Markov**: each field's token given the same field in the previous transaction.

If TxGPT beats the address-habit and Markov baselines, it has learned structure beyond counting. If it doesn't, that is reported plainly as a finding, and TxGPT is not chosen.

Scoring DS1 (`python -m lm.tx_score`):

1. Tokenise each DS1 address's full history with DS2's bucket edges (from `meta.json`).
2. Slide 768-token windows so that every transaction is scored with as much earlier context as fits (up to 127 earlier transactions).
3. A transaction's surprisal is $s_i = -\sum_{f \in \text{fields}} \log p(\text{token}_f \mid \text{everything before it})$, kept per field.
4. Two candidate scores: raw s_i, and s_i relative to the address (a robust z against the address's own earlier surprisals). The tuning plant set chooses between them.
5. Apply the same minimum-history rule as the baseline (at least 10 earlier transactions over at least 14 days), so the methods are compared on the same rows. The flag threshold is the top 2% of fit-period scores.
6. **Explanation.** The field with the largest surprisal names the reason: gap → "the timing was unusual for this address", amount → "the amount was unusual", inputs and outputs → "the transaction's shape was unusual", fee → "the fee rate was unusual".

Export. `models/txgpt.pt` (fp16 weights, about 30 MB for size L[est.]) plus its `meta.json`. Scoring runs on CPU in the marker's environment, within minutes for DS1.

10.2 The Explorer Assistant: a 152M text model

Stage 1, pretraining. The existing `configs/text_150m_pretrain.json`: 18 layers, $d = 768$, 12 heads, SwiGLU hidden 2,048, sequence 1,024, our 32,768-token BPE tokenizer. 2B tokens of DS5, subject to the gate in Section 5.3. Batch about 262,000 tokens; peak learning rate 6×10^{-4} falling to 6×10^{-5}; warm-up 500 steps; weight decay 0.1. It starts on the evening of 21 October.

Stage 2, chat fine-tune. The existing `configs/text_150m_chat.json`: 100M tokens of DS6. Loss only on assistant replies (the mask file); learning rate 2×10^{-4} falling to 2×10^{-5}; about 3–4 hours[est.].

Stage 3, explanation fine-tune (new). `configs/text_150m_explain.json`: start from the chat model; DS7 for 3 passes (about 8M tokens[est.]); learning rate 1×10^{-4}; warm-up 20 steps; loss on replies only; validation on DS7's validation split. Under an hour[est.].

Evaluation.

Measure	How
Language-modelling quality	Validation loss and perplexity on DS5's 10M held-out tokens after stage 1, on DS6 validation after stage 2, and on DS7 validation after stage 3. Training and validation curves from each run's `log.jsonl`.
Number faithfulness	On about 500 DS7 test inputs with greedy decoding: the share of outputs in which *every* number appears in the input, after formatting.
Limitation line	The share of outputs containing the fixed line exactly.
Banned words	The share of outputs containing any of: fraud, criminal, illicit, guilty, suspicious. The target is 0 after the validator.
Feature coverage	The share of contributing features the explanation mentions.
Similarity to the template	ROUGE-L against the template answer, written as a short LCS function rather than another dependency.
Human check	A teammate rates 30 random outputs for clarity and correctness (1–5), blind to which model wrote them; 10 template outputs are mixed in as controls.

Guardrail for A3. `explorer/assistant.py` runs the model, then validates the output: the numbers must match the input, the limitation line must be present, and no banned word may appear. If any check fails, it returns the template explanation instead. The pass rate is reported. This keeps the Explorer trustworthy even if the small model hallucinates, and the report says so: "the template is authoritative; the assistant only rephrases".

Honest framing for the report. These are *small* language models trained from scratch on one consumer GPU. They are not comparable to commercial LLMs. Their value is that they show the whole pipeline: tokenizer, pretraining, alignment, evaluation and a safety wrapper.

10.3 The Explorer Assistant 3B: a fine-tuned comparison (added 22 October)

The tutor, Arthur Hoang, agreed to an extra model: a pretrained base model of about 3B parameters (Qwen2.5-3B-Instruct, revision `aa8e7253`, chosen on 23 October: not gated, and already chat-tuned, so it can also serve as the Explorer's chatbot), fine-tuned by the team. It is added to the 152M Assistant and doesn't replace it.

Method. QLoRA: the base weights are loaded in 4-bit and only low-rank adapters are trained, which fits the RTX 3060's 12 GB. One stage: DS7 mixed with 2,500 DS6 conversations (so its general chat does not degrade), two passes, loss on replies only; about 10.5M tokens. The DS7 train, validation and test splits are the same ones the 152M model uses. A few hours of GPU time in total[est.].

When. After the 152M model's three stages and TxGPT are done (both have priority). It never pauses those runs.

Evaluation and guardrail. The same measures as the table above, on the same DS7 test inputs, so the report has a direct from-scratch vs fine-tuned comparison. It serves 4-bit quantised behind the same `explorer/assistant.py` validator, with the template answer as the fallback.

Records. The base model's name, revision and licence go in `LICENSES.md` and the model card. The change goes in `CHANGELOG.md` and the AI declaration. The adapter weights ship (or a link to them if they are too large); the base weights don't.

10.4 GPU timetable

When	GPU job
Wed 21 Oct, evening	Benchmark → apply the gate → start pretraining.
Thu 22 – Sun 25 Oct	Pretraining. Check loss and temperature twice a day.
Mon 26 Oct	Pretraining ends (or is paused if it overruns). TxGPT sweep, 9 runs.
Tue 27 Oct	TxGPT scoring of DS1; perplexity baselines. Resume pretraining if it was paused.
Wed 28 Oct	Chat fine-tune, 3–4 h.
Thu 29 – Fri 30 Oct	Build DS7 → explanation fine-tune → faithfulness evaluation. TxGPT plant evaluation (CPU or GPU).
After 30 Oct	3B QLoRA fine-tune (chat, then DS7) and its evaluation, a few hours. Then GPU free; re-runs only if a bug is found.

Watch out

If pretraining hasn't finished by Monday 26 October even after the gate, stop it there. Run stages 2 and 3 from the latest checkpoint and report the token count actually reached. TxGPT is the HD-critical model and must not wait for the text model.

Feeds the report and rubric

Report §6: two short paragraphs justifying TxGPT (sequence context, time-safe, explainable) and the Assistant (the tutor's requirement; an explanation layer behind a guardrail). Report §7.2: the perplexity table against the baselines, TxGPT's plant recall in the main comparison, and the Assistant's faithfulness table. Appendix: configs, training curves, the scaling plot and sample outputs. HD "additional ML": both models.

11 Workstream 5: evaluation

`python -m explorer.cli evaluate` runs everything in this section and writes each table to `reports/tables/` as CSV (plus a LaTeX version for the report). The report then quotes files, never numbers copied by hand.

11.1 On our own data (DS1): the A1 label-free plan, kept and extended

Every method enters the comparison: the baseline, K-Means, DBSCAN, regression (best model), Isolation Forest, LOF and TxGPT. Each check runs on the evaluation period only.

Check	Method	Measure	Target (A1)
Synthetic injection	50 plants per pattern × 5 reporting seeds (Section 7.4), scored by frozen models at each method's own threshold	Recall per pattern (mean ± SD); burst per transaction and per event; precision against planted labels as a lower bound; PR-AUC	≥70% for at least 2 of the 3 patterns
Stability	5 model seeds × 5 bootstrap resamples of the fit period; top 50 flagged on the evaluation period	Mean pairwise Jaccard	≥0.7

Check	Method	Measure	Target (A1)
Method agreement	Flag sets of all methods	Pairwise Jaccard heatmap; transactions only one method flags	Reported, no target
Reviewer check	Two teammates rate 20 top-flagged and 20 random unflagged transactions (selected method), shuffled and blind	Share of flagged rated "worth a look"; Cohen's κ	Reported as judgement, never as ground truth
Sanity rule	Flag share per address	Maximum share	No address above 20%
Flag rate	Share of evaluation transactions flagged	%	About 2% by design; reported

Notes on fairness.

- The baseline is deterministic, so only the bootstrap affects its stability, and that resamples addresses' fit periods, which the baseline doesn't use. Its stability is 1 by construction, and the report says so rather than presenting it as an achievement.
- Each method is compared at its own operating threshold (top 2% or $|z| \geq 3.5$), and also threshold-free with PR-AUC against planted labels. The second comparison stops a method from looking good simply because it flags more.
- Each plant must be detectable in principle. A spike landing where the address already sends 10× amounts is still counted. That kind of miss is part of the honest recall.

Needs a teammate

The reviewer check needs two people who did not build the scoring: Minh Khang Nguyen and Cam Dan Bui. On 29 October, `python -m explorer.cli reviewer-sheet` writes a shuffled CSV with 40 rows: transaction facts and the address's usual values, but no flag column. Each reviewer returns their ratings by 2 November. If one cannot do it, the report says the check was done by one reviewer, or not at all. It is never invented.

11.2 On the labelled data (DS3, DS4)

- **Main table**, one row per model × feature set (Elliptic) or model × split (BitcoinHeist): accuracy, precision, recall and F1 (positive class), macro-F1, PR-AUC, ROC-AUC.
- **Figures**: precision–recall curves (better than ROC under heavy imbalance; Saito and Rehmsmeier 2015), a confusion matrix for the best model, F1 per Elliptic test time step, and recall per held-out BitcoinHeist family.
- **The label check**: PR-AUC and precision/recall at the top 2% for each unsupervised method, against prevalence (chance) and the supervised random forest.

11.3 Language models

- TxGPT: test NLL per field against the three baselines; validation loss by size and seed; plant recall, stability and agreement as for every other method.
- Explorer Assistant: validation losses for the three stages, plus the faithfulness table and the guardrail pass rate (Section 10.2).

11.4 Choosing the method the Explorer uses

The A1 rule, applied as written: among methods with stability ≥0.7, choose the highest mean synthetic-injection recall. If two are within 5 percentage points, prefer the one whose flags are easier to explain. A1 left that order open; this plan fixes it as: baseline, regression interval, K-Means, Isolation Forest, DBSCAN, LOF, TxGPT. The chosen method becomes `explorer_method` in `settings.json`, and `predict.py` uses it. The report also states the runner-up, and whether combining two methods (flag if either flags) would raise recall without breaking the sanity rule. That combination is an option for A3, not a default.

11.5 Comparing results against the objectives

The report's evaluation section opens with this table, because the brief asks to "evaluate effectiveness by comparing results against objectives":

Objective (source)	Target	Result (filled from `reports/tables/`)
Snapshot built (A1 Obj. 1)	20–30 + 2–3 addresses	*n addresses, n transactions*
All methods score every transaction (A1 Obj. 2)	by 26 Oct	*date achieved*
Injection recall (A1 §2.3.2)	≥70% for 2 of 3 patterns	*per method*
Stability (A1 §2.3.2)	Jaccard ≥0.7	*per method*
Sanity (A1 §2.3.2)	≤20% per address	*max per method*
Labelled classification (A2 brief)	Report accuracy, precision, recall, F1	*best model per dataset*
TxGPT beats the habit baseline	lower test NLL	*yes or no, with values*
Assistant faithfulness	100% after guardrail	*raw and guarded rates*

11.6 Unexpected outcomes: planned for, not hoped against

The brief asks the report to "highlight unexpected outcomes". These are the most likely ones, each with what the report would say:

- **A method misses the 70% target.** This is A1 risk 4. Explain which pattern it misses and why (for example, a burst of typical amounts barely moves the size features), and fall back as the selection rule says.
- **Elliptic F1 drops after time step 43.** This is the drift Weber et al. (2019) describe. Show the per-step chart and discuss it as a known limit of static models.
- **BitcoinHeist leave-one-family-out recall is low.** Then the model recognises known families rather than ransomware behaviour in general. That is an important caution against deployment.
- **Unsupervised scores barely beat chance on Elliptic.** That is direct evidence for the Explorer's wording: unusual is not illicit.
- **TxGPT doesn't beat the address-habit baseline.** Then the data is too small or the tokens too coarse, and the report says so.
- **The refinements CR-02/04/05 don't help.** Then the method as written in A1 stays, and the evidence is reported.

Done when

Every table in this section exists as a CSV produced by one command. The selected method is recorded in `settings.json`. The objectives table is complete. Every figure the report uses is in `reports/figures/`. Results are frozen as version 1 on 29 October (M2). Later changes only

fix bugs, and each one is listed in the changelog.

12 Workstreams 6–9: prediction, code quality, README, packaging

12.1 Prediction interface

The brief's README requirement ends with "use your model for prediction". The A3 brief says the back end "runs predictions" on user input. One module serves both: `explorer/predict.py`. It loads saved artefacts once and returns plain dictionaries in the JSON shapes of the A1 prototype (`frontend/src/api/types.ts`).

Function	Returns
`score_address(address)`	The daily timeline, each transaction with its scores from every method, the selected method's flag, the contributing features, the comparison basis and the limitation line. Typed addresses outside the snapshot raise `Unsupported`; addresses with too little history raise `InsufficientHistory`. These match A1's planned 404 and 422 responses.
`score_what_if(address, amount_btc, direction, time_utc)`	The user enters a hypothetical next transaction for a snapshot address. Features are computed against the real history and every model scores it. This is the "user input → prediction" that A3's POST endpoint needs.
`classify_elliptic(rows)`	The probability and label (at the tuned threshold) from the best Elliptic model, for one or more feature rows.
`classify_heist(length, weight, count, looped, neighbors, income)`	The probability that an address-day resembles the labelled ransomware rows, worded exactly like that.
`txgpt_surprisal(address)`	Surprisal per transaction and per field.
`explain(tx_result, use_assistant=False)`	The template explanation, or the guarded Assistant text when `use_assistant=True`.

Command line (all shown in the README with their expected output):

```
python -m explorer.cli predict address bc1q...            # timeline + flags for a
    snapshot address
python -m explorer.cli predict what-if --address bc1q... --amount 0.05 --direction out
    --time 2026-10-01T12:00Z
python -m explorer.cli predict elliptic --rows data/processed/elliptic/example_rows.csv
python -m explorer.cli predict heist --length 18 --weight 0.008 --count 1 --looped 0 --
    neighbors 2 --income 1.2e8
python -m lm.chat --model models/assistant                  # chat with the Explorer
    Assistant
```

12.2 Code structure and comments

```
asm2/
  README.md, README.pdf          setup, rebuild, prediction, troubleshooting
  environment.yml                CPU environment (markers)
  environment-gpu.yml            CUDA environment (retraining language models)
  requirements.txt               pip fallback
  config/settings.json           every threshold, window, grid and path
  explorer/                      the pipeline package
    __init__.py  settings.py  cli.py
    esplora.py  amounts.py       API client and address amounts (from A1)
    snapshot.py  corpus.py       DS1 selection + fetch; DS2 crawl
```

```
    external.py                  DS3 / DS4 download, checksum, load
    transactions.py  features.py split.py  inject.py
    baseline.py  clustering.py  regression.py  outliers.py  classify.py
    evaluate.py  reviewer.py  predict.py  explain.py  assistant.py
    plots.py  tables.py  manifest.py
  lm/                            language models (existing + prepare_tx, tx_score,
    txgpt baselines)
  configs/                       text_150m_*.json, txgpt_{s,m,l}.json
  notebooks/01_eda.ipynb  02_models.ipynb  03_evaluation.ipynb
  data/  models/  reports/{figures,tables}/  tests/
  DATA_CARD.md  MODEL_CARD.md  LICENSES.md  SHA256SUMS.txt  CHANGELOG.md
```

Rules (code rubric lines 4 and 5):

1. One module per pipeline step, each runnable from `cli.py`. Notebooks only call package functions, so there is one implementation of everything.
2. Every module opens with a docstring saying what it does, its inputs and outputs, and which requirement or report section it serves (for example "DR-02").
3. Every public function has a docstring with arguments, units, return value and errors.
4. Comments explain *why*: why the window excludes same-block transactions, why the scaler sits inside the pipeline, why year is not a feature. Comments that merely restate the code are left out.
5. No magic numbers in code. Every threshold is read from `settings.json`, the A1 rule.
6. Type hints throughout; `pathlib` for paths, so they work on Windows; seeds passed explicitly; no global state except the loaded settings.
7. No dead code or commented-out experiments in the submission. Experiments live in tagged git history.

```
def robust_z(value: float, window: np.ndarray, min_scale: float, mad_scale: float =
    1.4826) -> float:
    """Modified z-score of `value` against `window` (Iglewicz and Hoaglin 1993).

    The spread has a floor (`min_scale`) so a history of identical values cannot divide
     by zero
    and turn every new value into an extreme flag (CR-05). `window` must hold only
    transactions
    confirmed strictly before the one being scored (DR-02); callers build it with
    `features.comparison_window`, never by slicing the full history.
    """
```

12.3 README

The README is worth 2 points: "comprehensive instructions for env setup, training, and prediction". It follows this outline exactly:

1. **What this is**, in three sentences, plus the folder map.
2. **Requirements**: Windows, macOS or Linux; Miniconda; 10 GB of disk; no GPU needed unless retraining the language models.
3. **Set up with conda**:

   ```
   conda env create -f environment.yml
   conda activate cos30049-a2
   python -m pytest -q                # expect: all tests pass in under 2 minutes
   ```

4. **Rebuild every result from the shipped data, offline**: `python -m explorer.cli all`

`-offline`. This runs the steps in order, with the expected time and output files listed for each: transactions → features → split → baseline → clustering → regression → outliers → classify (DS3, DS4) → TxGPT scoring → evaluate → figures.

5. **Each step on its own**, with the command, what it reads, what it writes, and how long it takes on a laptop.
6. **Fetching the data again** (optional, needs internet): `snapshot -refetch`, `corpus`, `external`. Explains the rate limits and that new data gives different numbers.
7. **Training**: scikit-learn models are trained by step 4. The language models are retrained with `environment-gpu.yml`, with the exact commands, configs and times measured on our RTX 3060.
8. **Prediction**: the commands from Section 12.1, each with its real output pasted in.
9. **Data**: a table of sources and licences, and a pointer to `DATA_CARD.md`.
10. **Troubleshooting**: `conda` not found; solver too slow (use `mamba` or the pip fallback); out of memory (`-n-jobs 2`); Kaggle token missing; Windows long paths; CUDA not detected.
11. **Reproducibility notes**: seeds, pinned versions, and the expected numeric tolerance across machines.

Needs a teammate

Milestone M4 (4 November): Le Quang Anh Hoang, who owns A1 risk 9, follows the README on a Windows machine that has never had the project, starting from a fresh download of the zip. They note every place they had to guess, and compare their `reports/tables/` with ours. Every guess becomes a README fix. A1's NFR-02 is met only when they finish without help.

12.4 Datasets and packaging

The inclusion rule is the brief's: data used by a final model, and nothing else.

Path in the zip	Contents	Size[est.]
`data/processed/esplora/transactions.csv`	DS1 clean table	1 MB
`data/processed/esplora/features.csv`	F1–F8, z-scores, scorable flag, period	2 MB
`data/processed/esplora/scores.csv`	Every method's score and flag per transaction	2 MB
`data/processed/esplora/injections/`	Plant specifications for every seed	1 MB
`data/processed/esplora/manifest.json`	DR-01 manifest	<1 MB
`data/raw/esplora/*.json.gz`	DS1 raw pages, as evidence for re-derivation	30 MB
`data/processed/corpus/ds2.parquet`	DS2 compact rows, hashed IDs	20 MB
`data/processed/corpus/tokens/`	TxGPT train, val, test token files and `meta.json`	6 MB
`data/processed/elliptic/`	Joined, split feature table (if the licence allows); otherwise `download.md` and checksums	0–80 MB

Path in the zip	Contents	Size[est.]
`data/processed/bitcoinheist/`	Processed rows used by the final models (training sample, test sets)	40 MB
`data/processed/ds7/`	Explanation pairs (train, val, test)	5 MB
`models/*.joblib` + cards	Every final scikit-learn model	30 MB
`models/txgpt.pt` + meta	TxGPT weights	30 MB
`models/assistant/`	152M weights (fp16, 305 MB): in the zip if Canvas allows; otherwise a link plus SHA-256, downloaded with one command given in the README	0 or 305 MB
`reports/`	Figures and tables	10 MB

Not in the zip, but recreatable: FineWeb-Edu (DS5) and smol-smoltalk (DS6), with scripts, dataset revisions and token-file checksums in `DATA_CARD.md`; the full BitcoinHeist CSV; the salt for DS2's address hashes.

Data card (`DATA_CARD.md`): for each dataset, the source and citation, licence, access date, collection method, size, every file and column with type and unit, missing values, known biases, and which model uses it. This is what earns "clear format" on the datasets line.

Model card (`MODEL_CARD.md`): one section per final model, covering its purpose, training data, parameters, metrics, limitations and intended use, plus the sentence "Unusual activity requires investigation; it does not indicate wrongdoing."

Size check, 22 October. Find Canvas's upload limit for this assignment, by asking the tutor or testing on a draft. The plan targets a zip under 200 MB, or under 500 MB if the 152M model ships in it. If the limit is lower, large items move to a link with a checksum, and the README says where.

Done when

The zip unpacks into the tree above. `SHA256SUMS.txt` verifies every file. A teammate rebuilds all of `reports/tables/` from it (M4). Every included file is used by a final model, and every final model's data is included or linked with a checksum.

13 Workstream 10: the report

The report is worth 12 points. It follows the brief's section list in the brief's order, so the completeness point can't be lost. The word budget is weighted by marks: data collection and processing carry 5 points, so they get the most words.

13.1 Cover and front matter (not counted)

- **Title**: "Bitcoin Address Activity Explorer: Detecting Unusual Transactions with Machine Learning". "Unusual", never "fraud".
- **Group identities**: group name HCMC_COS30049.1_4; group ID ("Group 4" on the contribution form); the four names and student IDs; **the tutor's name**; the unit; the submission date.
- The word count and how it was counted (Section 13.6).
- Table of contents, list of figures, list of tables.

13.2 Section-by-section plan

Section	Words	Content	Figures and tables
1 Introduction	170	Motivation: anyone who handles cryptocurrency must decide which of many transactions deserve a closer look, and an address's history is a long, unexplained list. Intended user: a student researcher examining one supported address (as in A1); analysts doing triage benefit too. What A2 delivers: data, models and evaluation that rank each transaction by how unusual it is for *its own address*, with a reason, plus labelled-data experiments showing what such scores can and cannot say. The limitation line, stated once.	–
2 Problem framing	210	The challenge: deciding which transactions deserve attention without trustworthy labels. Limits of existing solutions: block explorers show raw lists; fixed global rules ignore each address's scale; supervised models trained on Elliptic and BitcoinHeist rely on sparse, noisy labels and on anonymised or aggregated features that can't be computed for an arbitrary address (Weber et al. 2019; Elmougy and Liu 2023). Why ML fits: interactions between features, expected behaviour learned per address, structure found without labels, sequence context, and supervised learning where labels exist. Four research questions: RQ1, the best stable detector of planted patterns; RQ2, supervised performance on labelled data over time and on unseen families; RQ3, how far unusualness and labelled illicit activity overlap; RQ4, whether a from-scratch sequence model learns more than habit statistics.	–
3 Data collection	410	The sources table, then three short subsections. Our own collection (DS1, DS2): API, paging, pacing, the selection rule fixed before fetching, the verification sums, and the manifest. Public labelled data (DS3, DS4): citation, access, licence, what the labels mean. Text corpora (DS5–DS7). Challenges and responses, then the selection-bias statement (DR-02 condition 3).	T1 datasets; T2 challenges

Section	Words	Content	Figures and tables
4 Data processing	400	Cleaning steps with row counts; amount and direction; F1–F4 and the windows; the robust z; scorability; the time split; the leakage controls in 3–4 sentences; the Elliptic join and temporal split; BitcoinHeist binarisation, transforms, excluded leakage columns and sampling; TxGPT tokenisation; the injection design.	F1 pipeline; T3 cleaning log; T4 features
5 Data analysis	370	The five findings of Section 8, each written as observation, then statistic, then decision.	F2 distributions; F3 in vs out; F4 PCA density; F5 Elliptic balance over time
6 Model selection	400	The criteria C1–C6; the condensed candidates table; one paragraph each for clustering, regression, classification, the A1 extras and the two language models, saying why each was kept. The selection rule for the Explorer's method.	T5 candidates
7.1 Technical implementation	170	Libraries and why (pandas, scikit-learn, matplotlib, PyTorch for the language models); package structure and CLI; configuration in one file; seeds and pinned environments; hardware and measured run times; the prediction interface ready for A3.	(architecture is F1)
7.2 Implementation evaluation	560	The objectives table first. Then the DS1 comparison (recall by pattern, stability, sanity, agreement) and the selected method; the labelled results and the label check; the language-model results; unexpected outcomes; limitations; ethics in two sentences.	T6 objectives; T7 DS1 comparison; T8 labelled results; T9 LM summary; F6 recall; F7 agreement; F8 PR curves and F1 per step; F9 TxGPT
8 Conclusion	160	One answer per RQ; the method A3 will use and why; three limitations; future work (graph neural networks on Elliptic, OPTICS, more addresses, the A3 user study).	F10 example flagged transaction
Total	**2,850**	150 words of margin under the 3,000 limit.	

13.3 Appendices (not counted)

A. Data dictionary summary (the full one is in `DATA_CARD.md`).
B. Hyperparameter searches: every grid and the chosen values.
C. Full results: every model, split and seed.
D. Language models: configs, training curves, the TxGPT scaling plot, sample outputs.

E. Promises from A1 and how A2 met them (Section 2.4), plus a summary of CR-06.
F. Usability walkthrough results (A1 team action T7), only if the sessions happened.
G. Reproducibility: hardware, library versions, measured run times.

13.4 Writing rules

- Short sentences in plain English. The present tense describes the system; the past tense describes what was done. Each acronym is defined once.
- One term per concept, used everywhere: *unusual*, *flag*, *planted transaction*, *fit period*, *evaluation period*, *address history*.
- No banned word describes the tool or DS1. "Illicit" and "ransomware" appear only as the labelled datasets' own classes.
- Every number comes from `reports/tables/`, and a script checks the report's numbers against the CSVs before submission.
- Every figure and table is referenced in the text. Every caption says what to notice, not only what is shown.
- Claims about other work are cited. Claims about our own work point to a table or figure.

13.5 Referencing

Numbered Harvard, as in A1: an alphabetical, numbered list with author–date citations in the text. The draft list is Appendix C. Every entry is checked against its source before use: authors, year, title, venue, DOI or URL, and the access date for web sources. A small script lists citations without entries and entries never cited. Both lists must be empty.

13.6 Word count

`texcount` isn't installed, so `tools/wordcount.py` strips LaTeX from the body sections (Introduction to Conclusion) and counts the words. The brief excludes the cover, contents, bibliography and appendix, but says nothing about tables and captions.

Needs a teammate

Ask the tutor, in the CR-06 message, whether tables and captions count towards the 3,000 words. Until they answer, the plan keeps body text at or below 2,850 words and tables lean. The cover states the counting rule used, for example "2,840 words: body text excluding tables, captions, references and appendices".

13.7 How the report gets produced

1. LaTeX, reusing the A1 preamble. A1's preamble loads `xltabular`, `pdflscape`, `csquotes` and `seqsplit`, which the local TinyTeX doesn't have, so they are replaced with `longtable`, plain tables and `\textbackslash path` to make the build local and repeatable.
2. Figures are `\textbackslash includegraphics` of the PDFs in `reports/figures/`. Tables are `\textbackslash input` from LaTeX files generated next to the CSVs, so re-running the evaluation updates the report.
3. **Writing order.** Data sections on 1 November, once the data facts are frozen; models and evaluation on 2 November; introduction, problem framing and conclusion last, on 3 November, so they match the actual results.
4. **Review on 5 November.** Cam Dan Bui (report integrator in A1) reads it against the rubric and the completeness checklist and proofreads. Comments are resolved the same day.

13.8 Completeness checklist

Required by the brief	Check
Project title	☐
Group name and ID; student names and IDs; tutor's name	☐
Introduction: motivation and intended user	☐
Problem framing: challenge, limits of existing solutions, why ML	☐
Data collection: sources, methods, challenges and how they were addressed	☐
Data processing: missing values, normalisation, feature engineering, joining datasets	☐
Model selection: criteria, algorithms considered, rationale	☐
Technical implementation: libraries, high-level design, reasons for final decisions	☐
Implementation evaluation: results against objectives, performance, unexpected outcomes	☐
Conclusion: findings, insights, future work	☐
Bibliography in Harvard style	☐
Appendix	☐
At most 3,000 words; file named after the group; PDF	☐

14 Schedule

14.1 Day by day

Day numbers follow A1: day 0 is Monday 14 September 2026, and A2 is due on day 55.

Day	CPU and writing lane	GPU and network lanes	Teammates
Tue 20 Oct (36)	This plan. Send CR-06 and the tutor questions (Appendix A).	–	Approve CR-06 in Discord
Wed 21 Oct (37)	`git init`; copy `explorer` and `settings.json` across; set up both environments; CLI skeleton. DS1 selection, fetch and checks.	DS3 and DS4 downloads. FineWeb preparation → benchmark → gate → **start pretraining** overnight	–
Thu 22 Oct (38)	`transactions.py`, `features.py` (port), parity and time-safety tests, `split.py`.	DS2 discovery; **crawl overnight**. Pretraining continues	Cam Dan Bui: manual check of one transaction and the DS1 manifest review
Fri 23 Oct (39)	DS1 analysis (notebook 01); `inject.py` and tests.	DS2 crawl finishes, then its checks. Pretraining	–
Sat 24 Oct (40)	DS3 and DS4 processing and analysis; fix the DS4 split year. `prepare_tx` (DS2 tokens).	Pretraining	Le Quang Anh Hoang: code review of features, split and injection (24–25)

Day	CPU and writing lane	GPU and network lanes	Teammates
Sun 25 Oct (41)	Baseline and threshold sweep; Isolation Forest and LOF (tuning plants). **M1**.	Pretraining	–
Mon 26 Oct (42)	K-Means, DBSCAN, address profiles; regression (Ridge, random forest, gradient boosting, quantile).	Pretraining ends or is stopped → **TxGPT sweep**	–
Tue 27 Oct (43)	Elliptic models; search runs overnight.	TxGPT scoring and baselines	–
Wed 28 Oct (44)	BitcoinHeist models (3 splits); the label check on DS3 and DS4.	Chat fine-tune	–
Thu 29 Oct (45)	Evaluation suite; method selection; reviewer sheet sent out. **M2: results frozen v1**.	Build DS7 → explanation fine-tune	Minh Khang Nguyen and Cam Dan Bui: reviewer ratings, due 2 Nov
Fri 30 Oct (46)	`predict.py`, prediction CLI, Assistant guardrail; figures and LaTeX tables; data and model cards.	Assistant faithfulness evaluation	–
Sat 31 Oct (47)	Comment and docstring pass; notebooks 02 and 03; all tests green; README draft. **M3**.	GPU free	–
Sun 1 Nov (48)	Report §3–5. Dry run of the zip and its size.	–	–
Mon 2 Nov (49)	Report §6–7. Reviewer ratings in; κ computed.	–	–
Tue 3 Nov (50)	Report §1, 2 and 8; references; appendices; word count; number check.	–	–
Wed 4 Nov (51)	Fixes from the clean install; CPU run on the Mac; tag `v1.0`. **M4: code freeze**.	–	Le Quang Anh Hoang: clean install from the README
Thu 5 Nov (52)	Apply review comments; final figures; banned-word scan; draft the AI declaration.	–	Cam Dan Bui: rubric review and proofread
Fri 6 Nov (53)	Final PDF; build the zip from the tag; `SHA256SUMS`; unzip and run the quick check from the zip itself.	–	Team: minutes, contribution form
Sat 7 Nov (54)	**M5: submit.** Save the Canvas receipt; download what was submitted and open every file.	–	Leader submits
Sun 8 Nov (55)	Buffer only. Deadline 23:59 GMT+7.	–	–

14.2 Gantt chart

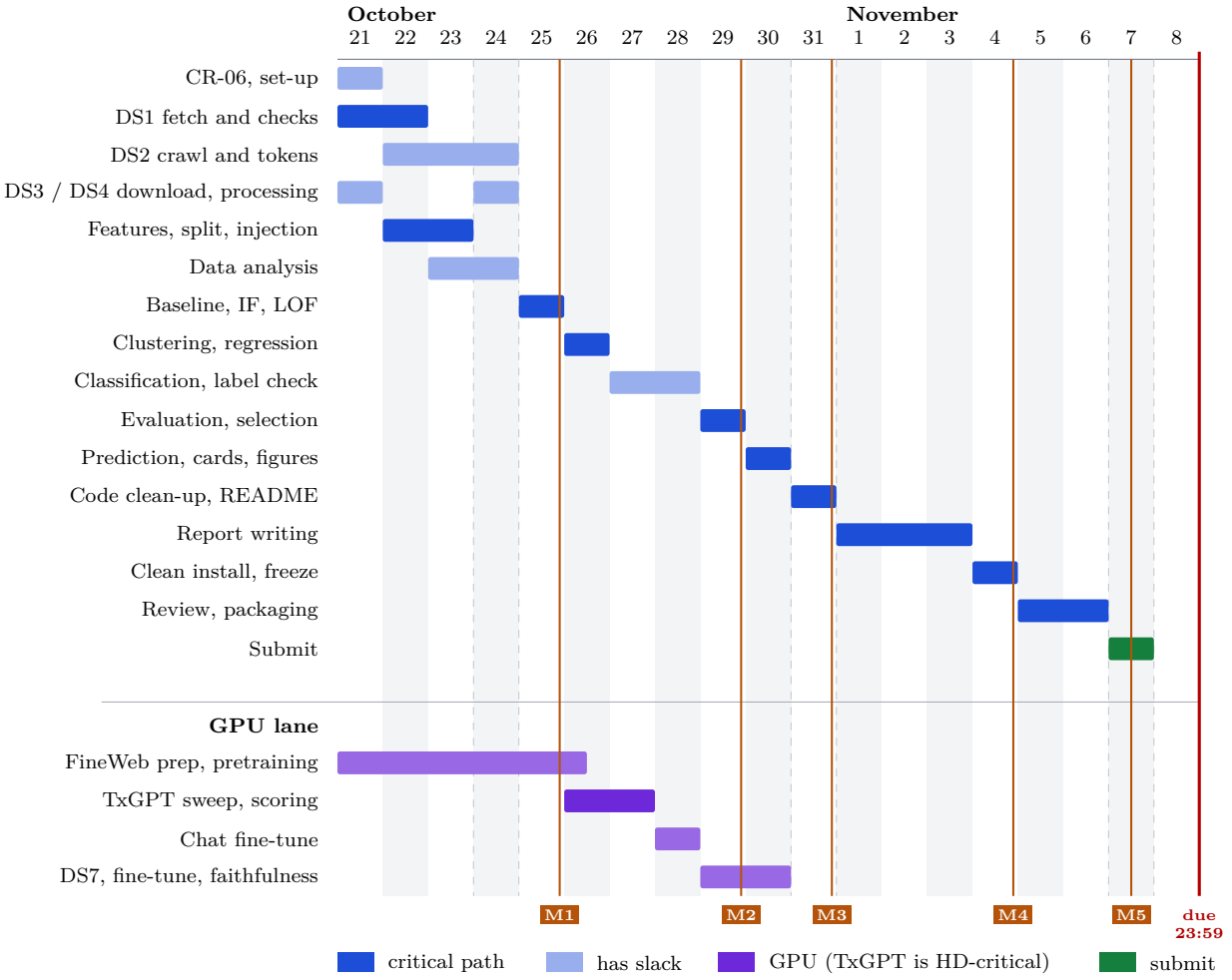


Figure 1: A2 schedule from Wednesday 21 October to Sunday 8 November. Dashed lines mark weekends; orange lines mark milestones M1–M5.

14.3 Milestones and what each one proves

ID	Date	Evidence that it happened
M1	Sun 25 Oct	`scores.csv` holds baseline, Isolation Forest and LOF scores for every scorable DS1 transaction (A1 Objective 2), and the DR-02 tests pass.
M2	Thu 29 Oct	Every evaluation table exists. The Explorer's method is selected and recorded. Results are tagged `results-v1`.
M3	Sat 31 Oct	Every planned function exists; notebooks run clean; tests pass; the README draft is complete.
M4	Wed 4 Nov	A teammate rebuilt `reports/tables/` from the README alone. The code is tagged `v1.0` and frozen; after this, only bug fixes go in, each logged.
M5	Sat 7 Nov	The Canvas receipt is saved, and the submitted files were downloaded back and opened.

14.4 The critical path and the slack

The critical path is: DS1 fetch → features and split → baseline → clustering and regression → evaluation → selection → prediction → clean-up → report → clean install → packaging → submission. Classification on DS3 and DS4 has about one day of slack, because the report's evaluation section doesn't need it until 2 November. The GPU lane has about three days of slack

before TxGPT's results are needed (29 October), and the text model's gate protects that slack. Sunday 8 November is a full spare day.

If a day slips. First use the slack in the non-critical rows. Then drop items from the bottom of the priority list (Section 15.3). Never touch the 7 November submission.

15 Risks, priorities and the submission checklist

15.1 Risk register for Assignment 2

The register continues A1's numbering, owners and scoring, so markers can follow a risk from one report to the next. A1 risks that still matter keep their number, with the trigger rewritten for A2. New risks start at 23. The score is likelihood × impact with A1's mapping: Unlikely 1, Low 2, Moderate 3, Likely and High 4; impact Low 2, Medium 3, High 4, Critical 5. Rows are sorted by score. The register is reviewed at every weekly meeting and after each milestone, and the minutes record any change.

No.	Risk	L × I	Trigger / owner	Prevention; fallback
4	Methods fail the label-free checks (plant recall or stability)	High × High 4×4 = 16	Recall below 70% for 2 of 3 patterns, or Jaccard below 0.7, at M2 (29 Oct) / Cao Tri Tran	*Prevention:* threshold sweep and separate tuning plants; the CR-02/04/05 ablation; seven candidate methods instead of three. *Fallback:* the selection rule picks the best stable method, even if that is the baseline; the report explains the miss (Section 11).
9	Code doesn't run on the marker's machine	Likely × High 4×4 = 16	The M4 clean install fails or needs help / Le Quang Anh Hoang	*Prevention:* CPU-only `environment.yml` with pinned versions; `requirements.txt`; offline rebuild; `pathlib` paths; tests as the quick check. *Fallback:* README troubleshooting with expected output pasted in; contact the tutor at once.
7	Critical-path tasks slip	Moderate × Critical 3×5 = 15	A critical-path row ends a day late, or DS1 isn't fetched by the end of 21 Oct / Le Quang Anh Hoang	*Prevention:* DS1 starts on day 37; a daily check against Section 14; the GPU lane is off the critical path. *Fallback:* use slack, then cut from the bottom of the priority list (Section 15.3). The 7 November submission never moves.
6	Scope creep: HD extras eat time that Must items need	Likely × Medium 4×3 = 12	Work on a Should or Could item while a Must item is behind / Le Quang Anh Hoang	*Prevention:* the priority list; the pretraining gate; GPU jobs run unattended. *Fallback:* stop the item and record the cut in the minutes.

No.	Risk	L × I	Trigger / owner	Prevention; fallback
13	Unsupported claims about real transactions	Moderate × High 3×4 = 12	Reworded by D-A2-3: any draft claims precision, recall, F1 or accuracy for real snapshot transactions / Cao Tri Tran	*Prevention:* the metrics rule (Section 4.3); "against the dataset's labels" wording; banned-word scan; number-check script. *Fallback:* delete the claim and restate it as a ranking with limitations.
23	Pretraining is slower than estimated, or is interrupted	Likely × Medium 4×3 = 12	The benchmark projects more than 72 h for 2B tokens, or a run stops overnight / Cao Tri Tran	*Prevention:* the gate (Section 5.3); checkpoints every 30 minutes; Windows Update paused; resume already tested. *Fallback:* cut tokens to the measured budget (floor 1B); below the floor, TxGPT is the only language model, as CR-06 records.
26	The small text model writes wrong numbers or banned words	High × Medium 4×3 = 12	Raw number faithfulness below 95% on the DS7 test set / Cao Tri Tran	*Prevention:* the guardrail; the template is authoritative; five phrasings per template in DS7. *Fallback:* `use_assistant=False` stays the default; the report gives raw and guarded rates.
30	A bug found after the results freeze changes numbers	Moderate × High 3×4 = 12	A test failure or review finding after M2 / Cao Tri Tran	*Prevention:* tests green before M2; tables generated, never typed; report numbers checked by script. *Fallback:* fix, re-run `all -offline` (under an hour on CPU), regenerate tables, log it in `CHANGELOG.md`.
31	Generative-AI use undeclared or under-declared	Low × Critical 2×5 = 10	No declaration drafted by 5 Nov / Cam Dan Bui	*Prevention:* the declaration is a listed submission file; a running log of what AI helped with, kept from today. *Fallback:* ask the tutor how to add it if it was missed.
2	A dataset is unavailable or its licence is unclear	Moderate × Medium 3×3 = 9	Elliptic's Kaggle licence not confirmed by 24 Oct, or a download fails / Cao Tri Tran	*Prevention:* download on 21 Oct; record each licence; ship a script and checksums when redistribution is unclear. *Fallback:* BitcoinHeist (CC BY 4.0) carries the classification alone; the Elliptic++ transaction set is the substitute.
5	A teammate can't do their review task in time	Moderate × Medium 3×3 = 9	Manifest review (22 Oct), code review (24–25 Oct), ratings (2 Nov), clean install (4 Nov) or proofread (5 Nov) late / Le Quang Anh Hoang	*Prevention:* every task has a name and a date in the CR-06 message, plus a named backup. *Fallback:* the backup does it; otherwise the report says who did the check, or that it was not done.

No.	Risk	L × I	Trigger / owner	Prevention; fallback
15	Esplora rate limits block DS1 or DS2	Moderate × Medium 3×3 = 9	More than 5% of requests return 429 in an hour, or DS2 has under 3,000 addresses by 23 Oct / Le Quang Anh Hoang	*Prevention:* pacing, back-off, caching, a resumable state file, the mempool.space fallback, overnight crawling. *Fallback:* accept the 3,000 floor and say so; a second crawl night uses DS2's slack.
19	Sample selection bias	Moderate × Medium 3×3 = 9	No written rule in the manifest before fetching / Cao Tri Tran	*Prevention:* the rule is fixed and seeded first, and every skip is logged. *Fallback:* a limitations paragraph states the bias.
22	The prototype promises what A2 can't deliver	Moderate × Medium 3×3 = 9	`predict.py` output differs from the shapes in `types.ts` / Minh Khang Nguyen	*Prevention:* `test_predict.py` checks every shape. *Fallback:* update the prototype through change control.
24	Out of memory or disk	Moderate × Medium 3×3 = 9	Under 40 GB free on 21 Oct, swapping during random-forest tuning, or CUDA out of memory / Cao Tri Tran	*Prevention:* memory-mapped tokens; `-delete-raw`; capped `n_jobs`; heavy CPU jobs never overlap. *Fallback:* micro-batch 4 with more accumulation (same effective batch); `n_jobs=4`; raw shards on an external disk.
27	TxGPT doesn't beat the habit or Markov baselines	Moderate × Medium 3×3 = 9	Test NLL no lower than the address-habit baseline at M2 / Cao Tri Tran	*Prevention:* the size sweep with 3 seeds; DS2 stretch target of 10,000 addresses. *Fallback:* report it as a finding. It is still a valid experiment, but TxGPT is then not the Explorer's method.
28	The zip exceeds Canvas's upload limit	Moderate × Medium 3×3 = 9	The 1 Nov dry-run zip is over the confirmed limit / Le Quang Anh Hoang	*Prevention:* the limit checked on 22 Oct; DS2 in Parquet; fp16 weights. *Fallback:* the Assistant weights and Elliptic move to a link with SHA-256, and the README says where.
29	The report goes over 3,000 words, or tables count	Moderate × Medium 3×3 = 9	Body above 2,850 words on 3 Nov, or the tutor says tables count / Cam Dan Bui	*Prevention:* the word budget per section (Section 13); lean tables; detail in appendices. *Fallback:* move tables to appendices; trim §7.1 and §2 first.
14	Data leakage inflates results	Low × High 2×4 = 8	Results unexpectedly strong (Elliptic far above Weber et al. 2019; random split far above the temporal split), or a DR-02 test fails / Cao Tri Tran	*Prevention:* the leakage controls (Section 7.8) and tests; the test period is used once. *Fallback:* rebuild, re-run, replace the numbers, log it.

No.	Risk	L × I	Trigger / owner	Prevention; fallback
16	Real addresses read as accusations, or terms forbid a use	Low × High 2×4 = 8	Any text implies wrongdoing by a DS1 or DS2 address, or a licence forbids the planned use / Le Quang Anh Hoang	*Prevention:* hashed DS2 IDs with the salt kept out of the zip; the limitation line; no labels attached to real addresses; licences recorded. *Fallback:* shorten or replace IDs; drop the dataset.
25	CR-06 is rejected or the answer is delayed	Low × High 2×4 = 8	No tutor reply by 23 Oct, or a "no" / Cam Dan Bui	*Prevention:* sent on 20–21 Oct; the plan builds every reading (all three types, IF and LOF, both language models), so almost any answer is already covered. *Fallback:* continue on the plan's reading, which meets the brief's literal text; record the open question in the minutes; adjust only the report's framing.
10	Upload or naming error causes a late penalty	Unlikely × Medium 1×3 = 3	Not submitted 24 hours before the deadline / Le Quang Anh Hoang	*Prevention:* the checklist in Section 15.4; submit on 7 Nov. *Fallback:* resubmit at once; if past the deadline, tell the tutor and submit anyway.

A1 risks not carried forward: 1 (now covered by 4 and the threshold sweep), 3 (an A3 usability question), 8 (A1's demo video), 11 (the team handles minutes separately), 12 (covered by 13 and 16), 17 (D1 passed), 18 (the amount derivation is tested; still checked in DS1's sums), 20 (resolved) and 21 (the Confluence trial ended on 15 October, so the change log now lives in the repository as `docs/change-log.md`).

15.2 Early-warning signals

Most of these risks show up early if someone looks. Each signal below takes under a minute to check, and the morning check runs every day from 21 October to 7 November.

Signal	Threshold	Action
GPU tokens per second against the benchmark	15% lower	Check temperature and other GPU use; restart from the checkpoint if speed doesn't recover (risk 23).
Pretraining validation loss after warm-up	No fall over 3 evaluations	Inspect `log.jsonl`; check the learning rate and the data order before spending more GPU hours.
GPU temperature	Above 83 °C	Improve airflow; lower the power limit in the NVIDIA settings if needed.
Crawl 429 rate	Above 5% over an hour	Slow the pace to 1 s; switch to mempool.space for an hour (risk 15).
Free disk space	Below 15 GB	Delete used FineWeb shards and old checkpoints, keeping `ckpt_last.pt` and `model_best.pt` (risk 24).
Critical-path row	Not done by the end of its day	Re-plan the next morning; cut from the priority list (risk 7).
Test suite	Any failure	Stop new features until it is green again (risk 30).
Section word count while drafting	Over its budget	Trim the same day (risk 29).

15.3 Priority list

The list decides what gets cut if time runs short. **Must** items give a complete submission and keep A1's promises; they are never cut, only shrunk (fewer grid points, a smaller DS2). **Should** items earn the HD points. **Could** items are done only with spare time. **Won't** items are named so nobody starts them.

ID	Item	Earns
Must		
M-1	DS1 snapshot, checks and manifest	Report 1; Code 2; A1 Obj. 1
M-2	Cleaning, F1–F4, the time split and the DR-02 tests	Report 1; Code 1
M-3	Baseline, Isolation Forest and LOF scoring every transaction	A1 Obj. 2; Code 1
M-4	K-Means and DBSCAN; regression models	Brief's model types; Code 1
M-5	Classification on DS3 (on DS4 if DS3's licence blocks it)	Brief's metrics; Code 1
M-6	Evaluation suite, method selection, objectives table	Code 1; Report 6
M-7	`predict.py` and the prediction CLI	Code 3
M-8	README, docstrings, package structure, data card	Code 2–5
M-9	The report, the zip and the AI declaration	Report 1–6
Should (if cuts are needed, cut from the bottom of this list up)		
S-1	DS2 crawl and TxGPT (the tutor's language-model requirement)	HD ML; HD datasets
S-2	DS4 BitcoinHeist, with all three split schemes	HD datasets
S-3	The label check on DS3 and DS4	HD ML; Report 2
S-4	Reviewer check (A1 promise; depends on two teammates)	Report 7.2
S-5	Text-model pretraining and chat fine-tune	HD ML; HD datasets
S-6	DS7, the explanation fine-tune and the guardrail	HD ML; A3
S-7	Address-profile clustering on DS1 and DS2	Report 2; A3
S-8	A model card per final model	Code 2
Could		
C-1	DS9: score Ransomwhere addresses with our own pipeline	HD datasets
C-2	DS8: address-level classification on Elliptic++ actors	HD datasets
C-3	Multi-class family classification on DS4	HD ML
C-4	"Flag if either method flags" as an option for A3	A3
C-5	A TxGPT ablation with finer amount buckets	HD ML
Won't (in A2)		
W-1	Graph neural networks on Elliptic	future work
W-2	Live API calls from the Explorer	A1 scope rule
W-3	Fine-tuning a public pretrained model	tutor's rule
W-4	Ethereum or other chains; XGBoost or LightGBM	scope; dependencies
W-5	Whole-chain datasets (ORBITAAL, the Bitcoin temporal graph)	16 GB RAM; schedule

Cut order if a day is lost: C-5, C-4, C-3, C-2, C-1, then S-8, S-7, S-6, S-5. Cutting S-5 leaves TxGPT as the only language model, which the tutor question in CR-06 prepares for. S-4 is cut only if the teammates can't do it. S-1 is cut last because it is the HD-critical model.

15.4 Submission checklist

Run on 6 November (build) and 7 November (submit). Every line is ticked by the named person, and the finished list is pasted into that week's minutes.

Check	Who	When
Report PDF		
Named `HCMC_COS30049.1_4_Assignment2_Report.pdf`; opens; fonts embedded	Cao Tri Tran	6 Nov
Cover has the group name and ID, four names and IDs, **the tutor's name**, the date and the word count	Cam Dan Bui	5 Nov
Body at most 3,000 words (`tools/wordcount.py`); the counting rule is stated	Cao Tri Tran	3 and 6 Nov
Every section the brief requires is present (Section 13 checklist)	Cam Dan Bui	5 Nov
Every figure and table is referenced in the text; captions say what to notice	Cam Dan Bui	5 Nov
Citation check clean: nothing cited but unlisted, nothing listed but uncited	Cao Tri Tran	6 Nov
Number check clean: every number matches `reports/tables/`	Cao Tri Tran	6 Nov
Banned-word scan clean for the tool and DS1	Cao Tri Tran	6 Nov
Code zip		
Built from tag `v1.0` by the packaging script	Cao Tri Tran	6 Nov
`SHA256SUMS.txt` verifies every file	Cao Tri Tran	6 Nov
Unzipped into an empty folder: `python -m pytest -q` passes and one prediction command runs	Le Quang Anh Hoang	6 Nov
Contains README (`.md` and `.pdf`), `DATA_CARD.md`, `MODEL_CARD.md`, `LICENSES.md`	Cao Tri Tran	6 Nov
Contains no salt, `.git`, caches, raw FineWeb or unused data	Cao Tri Tran	6 Nov
Size under Canvas's limit; anything linked has a SHA-256 in the README	Le Quang Anh Hoang	6 Nov
Prepared by the team separately		
Meeting minutes for every week since the team formed (W1–W8), including CR-06's approval and the reviews	Cam Dan Bui	6 Nov
Contribution form signed by all four members	All	6 Nov
Declaration and upload		
AI declaration: what AI helped with, and that the team checked and owns the work	Cam Dan Bui	5 Nov
All five files uploaded on Canvas by one member	Le Quang Anh Hoang	7 Nov
Canvas receipt saved; the submitted files downloaded back and opened	Le Quang Anh Hoang	7 Nov

15.5 After submission: what Assignment 3 inherits

- The tag `v1.0` and the frozen `reports/tables/`. A3 builds on them without changing A2's numbers.
- `explorer/predict.py` and its JSON shapes, which the FastAPI back end wraps directly.
- The guarded Explorer Assistant, which is optional in A3 and off by default.
- The usability walkthrough (A1 team action T7) and the A3 user study, which reuse A1's observer sheet.
- A short lessons-learned note in the Week 9 minutes: what took longer than planned, and why.

A Change request CR-06 and the messages to send

A1 §8.5 requires a change request to change a must-have item, add or remove a work package, change acceptance criteria, or use more than half of an assignment buffer. CR-06 does all four, so it needs every member's approval. It also touches the brief's marking requirements, so the tutor sees it before work depends on it.

A.1 Change log entry

The fields follow A1's Change Log (ID, date, requestor, proposal, impact, decision, approvers, changed artifacts). The Confluence trial ended on 15 October (A1 risk 21), so from CR-06 onwards the log lives in the repository as `docs/change-log.md`, with CR-01 to CR-05 copied in from the exported CSV.

Field	Entry
ID	CR-06
Date raised	Tuesday 20 October 2026
Requestor	Cao Tri Tran (Data/ML Lead)
Proposal	Adopt decisions D-A2-1 to D-A2-5 (Section 4): (1) cover clustering, regression and classification, keeping the baseline, Isolation Forest and LOF as extras; (2) add datasets DS2–DS7, with DS8 and DS9 optional; (3) report label-based metrics only where labels exist; (4) train two language models from scratch; (5) one repository with a CPU environment for markers and a GPU environment for retraining.
Reason	The A2 brief requires at least two of three named model types and reports accuracy, precision, recall and F1. A1's two outlier detectors may not count as two types, and A1's single unlabelled dataset can't support those metrics honestly. The HD band rewards additional datasets and additional ML implementations, and the tutor asked for a locally trained language model.
Impact on scope	Must items are added (clustering, regression, classification, prediction interface). Nothing A1 promised is removed: every A1 method and check is still delivered (Section 2.4).
Impact on schedule	A1's baseline put WBS 1.2.1 on 11–13 October; it now starts on 21 October. The new schedule (Section 14) runs four lanes in parallel and still submits a day early, but it uses more than half of the A2 buffer.
Impact on risk	Risk 13's trigger is reworded (Section 4.3). New risks 23–31 are added (Section 15.1).
Impact on quality	More evidence: three labelled-data experiments, a reviewer check, a clean-install test and two model cards.
Cost	No money. About 70–90 hours of GPU time on the Data/ML Lead's own PC (Section 5), and its electricity.
Decision	*Pending*: approved, approved with changes, or rejected.
Approvers	Cam Dan Bui; Cao Tri Tran; Le Quang Anh Hoang; Minh Khang Nguyen. The tutor is informed.
Changed artifacts	WBS and WBS dictionary (1.2.1–1.2.8), schedule baseline (v2), risk register, requirements DR-01 and DR-02 (wording only), `settings.json`.

A.2 Effect on the A1 work breakdown

Durations are A1's planned working days. The owners stay as A1 named them (Cam Dan Bui and Cao Tri Tran for every 1.2 package), except the new 1.2.8.

Package (A1 code)	A1 days	Change under CR-06
1.2.1 Build data snapshot	3	Also: the DS2 crawl, the DS3 and DS4 downloads, and DS5–DS6 preparation. DS1 itself is unchanged.
1.2.2 Clean data and build features	4	Also: the DS3 join and temporal split, DS4 transforms and sampling, TxGPT tokenisation, the refinement ablation.
1.2.3 Explore and visualise	3	Also: the labelled-data analyses and the DS2 checks (Section 8).
1.2.4 Build baseline and models	5	Now all three model types plus the A1 extras (Section 9).
1.2.5 Evaluate and select	3	Also: labelled-data metrics, the label check and the language-model evaluation. A1's selection rule is unchanged.
1.2.6 Code, README and zip	2	Also: `predict.py`, the data and model cards, and the M4 clean install.
1.2.7 Write report	4	Same 3,000-word limit; the content follows Section 13.
1.2.8 Language models (new)	–	TxGPT and the Explorer Assistant on the GPU lane, in parallel with 1.2.2–1.2.5, about 3 days of float. Owner: Cao Tri Tran. Done when both models are trained, evaluated and exported (Section 10).

A.3 Approval steps

1. **20 October.** Post the team message below in Discord, with this plan's PDF attached.
2. **By 21 October, 12:00.** Each member replies "approve" or states an objection. An objection is discussed in a call that evening; nothing waits longer than that.
3. **21 October.** Send the tutor message below. Work that doesn't depend on the answers (DS1, features, the baseline) carries on meanwhile.
4. Record the decision, approvers and date in `docs/change-log.md`, and save the schedule as baseline v2.
5. The Week 6 minutes record the approval with names and dates, and the tutor's answers when they arrive.

A.4 Message to the tutor

Subject: COS30049 HCMC_COS30049.1_4: Assignment 2 approach and four quick questions

Dear *<tutor's name>*,

Our Assignment 1 plan used outlier detectors (Isolation Forest and LOF) on an unlabelled Bitcoin snapshot. To meet the Assignment 2 brief, we plan to:

- add clustering (K-Means, DBSCAN) and regression on our own snapshot, and classification on two labelled public datasets (Elliptic and BitcoinHeist), keeping Isolation Forest and LOF as additional experiments;
- report accuracy, precision, recall and F1 only against those datasets' own labels, never for our unlabelled snapshot;
- train two language models from scratch on our own GPU: TxGPT, a small transformer over sequences of Bitcoin transactions, and a 152M-parameter text model that explains flags in plain language. Neither starts from anyone else's weights.

Could you confirm four things?

1. Does this mapping meet "at least two types of machine learning models"?
2. Does TxGPT on its own satisfy the language-model requirement for HD, or do you expect a text model as well?
3. Do tables and captions count towards the 3,000 words?

4. What is the Canvas upload limit for the zip? If the text model's weights (about 305 MB) are too large, may we link them with a checksum instead? The same question applies to the Elliptic data if its licence doesn't allow redistribution.

We will include a generative-AI declaration as you asked. If there is a template, we will use it.

Kind regards,
Cao Tri Tran, for HCMC_COS30049.1_4

A.5 Message to the team

CR-06 (A2 plan): please approve by Wednesday 12:00. Full plan attached. In short: we add clustering, regression and classification so A2 meets the brief's model-type rule; we add labelled datasets (Elliptic, BitcoinHeist) for honest precision and recall; and I train two language models from scratch on my GPU for HD. Every A1 promise is still delivered.

What I need from each of you, with dates:

- **Cam Dan Bui:** DS1 manifest review and one manual transaction check (22 Oct); reviewer ratings (29 Oct to 2 Nov); rubric review and proofread (5 Nov); the AI declaration draft (5 Nov). Also: what is our tutor's full name for the cover?
- **Le Quang Anh Hoang:** code review of `features.py`, `split.py` and `inject.py` (24–25 Oct); clean install from the README on a fresh Windows machine (4 Nov); Canvas upload (7 Nov).
- **Minh Khang Nguyen:** reviewer ratings (29 Oct to 2 Nov); check that `predict.py`'s output still fits the prototype's screens (30 Oct).
- **Everyone:** minutes for Weeks 1–8 and the contribution form, by 6 Nov.

Reply "approve" or tell me what to change.

B Datasets considered

The HD criterion rewards a "large number of additional datasets", but only datasets that a final model uses go in the zip, and each one costs collection, processing and report words. So the candidates were checked against four tests: is it Bitcoin, does it carry labels we can use honestly, is its licence clear, and does it fit a 16 GB machine and a 19-day schedule. The datasets in use are DS1–DS9 (Section 4.2). The table below lists the eighteen others, checked on 20 October 2026, with the decision for each. It also repeats DS9, the optional Ransomwhere set, so its reasoning sits beside the rest.

Decisions: *Fallback* means it replaces a chosen dataset if that one fails; *Could* means it is on the priority list's Could tier; *Not used* means it fails one of the four tests; *Won't* means it is ruled out for A2 (Section 15.3).

Dataset (source)	What it holds	Licence and access	Decision and reason
Bitcoin, labelled			
Elliptic++ transactions (Elmougy and Liu 2023)	Elliptic's transactions with 183 features and the same labels; also wallet-level (actor) files with 56 features	None stated; Google Drive link from the GitHub README	*Fallback* for DS3 if Elliptic's Kaggle licence blocks it. The actor files are DS8.

Dataset (source)	What it holds	Licence and access	Decision and reason
Elliptic2 (Bellei et al. 2024)	122,000 labelled subgraphs inside a background graph of 49M node clusters and 196M edges	Code Apache 2.0; data licence not stated; download via Elliptic's website	*Not used.* Subgraph labels need graph neural networks (W-1), and the background graph is too large for 16 GB of RAM.
BABD-13 (Xiang et al. 2024)	544,462 labelled addresses of 13 behaviour types, 148 features, July 2019 to May 2021	Host and licence not confirmed	*Not used* unless a confirmed copy appears. It would be a third classification set with little new to say, and its features need a subgraph tool we would have to rebuild.
Ransomwhere (Cable 2024)	Crowdsourced ransomware payment addresses with transactions and family; 5.6 MB JSON	CC BY 4.0; Zenodo	*Could*, as DS9. Its addresses run through our own pipeline, which gives a label check in our own feature space.
Showing the Receipts (GitHub, `cablej`)	1,013 likely ransomware payment addresses	CC BY 4.0	*Not used.* Same purpose as Ransomwhere but smaller; kept as a cross-check if DS9 runs.
Ransomware seed addresses (Paquet-Clouston, Haslhofer and Dupont 2019)	Seed address CSV, with an expanded set on Zenodo	No licence file	*Not used.* No licence, and it overlaps DS4's label sources.
OFAC sanctioned addresses (GitHub, `0xB10C`)	Bitcoin addresses on the US sanctions list, regenerated nightly	Repository licence not stated	*Won't.* A "sanctioned" label on real addresses is an identity claim, which A1 risk 16 rules out.
Bitcoin Ponzi addresses (Bartoletti, Pes and Serusi 2018)	Addresses of Ponzi schemes with features	Host and size not found	*Not used.* No confirmed download.
WalletExplorer entity labels	Exchanges, pools and services found by co-spend clustering	No bulk download or licence	*Not used.* No licence; research reports stale names.
Bitcoin, whole chain or graph			
ORBITAAL (Coquidé and Cazabet 2024)	Entity-level temporal graph, January 2009 to January 2021	CC BY 4.0; Zenodo; 156.9 GB	*Won't* (W-5): size.
Bitcoin temporal graph (Schnoering and Vazirgiannis 2024)	252M entity nodes, 786M edges, 34,098 labelled entities	CC BY 4.0; Figshare; PostgreSQL, about 120 GB	*Won't* (W-5): size. Good future work for address profiles.
BigQuery `crypto_bitcoin` (Google public dataset)	The whole chain as SQL tables	First 1 TB of queries a month free; needs a Google Cloud project	*Fallback* for the DS2 crawl if Esplora blocks it.
Blockchair database dumps	Blocks, transactions and outputs as TSV	Licence not confirmed	*Not used.* Licence unclear; Esplora covers the need.

Dataset (source)	What it holds	Licence and access	Decision and reason
SNAP Bitcoin Alpha (Kumar et al. 2016)	3,783 traders, 24,186 trust ratings from −10 to +10	Public download from SNAP	*Not used.* Trust ratings between traders, not transactions.
Not Bitcoin			
IBM AML (Altman et al. 2023)	Synthetic bank transfers with an "Is Laundering" label, in six sizes	Kaggle; licence not confirmed	*Not used.* Bank transfers, not Bitcoin.
SAML-D (Oztas et al. 2023)	About 9.5M synthetic monitored transactions, 12 features, 28 typologies	Kaggle; licence not confirmed	*Not used.* Same reason.
SynthAML (Jensen et al. 2023)	20,000 bank alerts (17% reported) and over 16M transactions	Figshare; data licence not explicit	*Not used.* Same reason.
Ethereum fraud sets (Kaggle, XBlock, Forta)	Labelled Ethereum accounts, for example 9,841 accounts with 22% flagged	Varies	*Won't* (W-4): outside A1's Bitcoin scope.
Card-fraud sets (ULB credit card, PaySim, Sparkov)	Card or mobile payments with fraud labels	Varies	*Not used.* Not Bitcoin.

How the report uses this. One sentence in the data-collection section, "Eighteen other public datasets were assessed against four criteria (Appendix X)", plus this table condensed in the report's appendix. That shows selection, which the HD band and data criterion reward, at no cost to the word limit. Entries that the report names must be cited (the second list in Appendix C). Details marked "not confirmed" are checked again before they appear in the report.

C Draft reference list

This is the starting list for the report's bibliography, in A1's numbered Harvard style: alphabetical, numbered, with author–date citations in the text. The report keeps only the entries it actually cites, and the citation script (Section 13) checks that.

How far each entry is checked. † means the details were confirmed on the source page on 20 October 2026. ‡ means the entry is copied from A1's reference list, which was checked for A1. Unmarked entries were written from the published record and must be checked against their DOI or source page before the report uses them. Entries with more than four authors are shortened to "et al."; expand them if Swinburne's Harvard guide asks for every author.

Main list

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Letters for same-author, same-year entries. When the final list is built, entries with the same author and "n.d." get letters in title order, continuing A1's sequence (A1 used Li n.d.a–e and Swinburne n.d.a–c). The citation script assigns them so the text and the list always agree.

Second list: only if the report names them

Datasets that were considered (Appendix B) or are optional. Each goes into the main list only if the report cites it.

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D Command reference, in run order

Every command in this plan, in the order it runs, from the repository root in an Anaconda Prompt on the training PC. Commands under `lm.` that exist today are marked *existing*; the rest are what this plan builds, with the names used throughout the plan. Each `explorer.cli` command prints what it read, what it wrote and how long it took, and appends that to `reports/logs/<step>.log`, so the report's processing numbers come from logs rather than memory.

D.1 Day 1 (21 October): set-up and checks

```
git init                                        # in asm2/, then copy explorer/ and
    config/ from asm1_demo
conda env create -f environment-gpu.yml         # today's environment.yml, renamed; env:
     cos30049-a2-gpu
conda env create -f environment.yml             # new CPU-only file for markers; env:
    cos30049-a2
conda activate cos30049-a2-gpu
nvidia-smi                                      # driver and CUDA version
python -c "import torch; print(torch.cuda.is_available(), torch.cuda.get_device_name(0)
    )"
python -m pytest -q                             # existing 15 LM tests + 9 amount tests
python -m lm.smoke                              # existing: expect 8/8 stages passed
```

D.2 Data collection (Section 6)

```
python -m explorer.cli snapshot                 # DS1: select (seed 30049), fetch,
    verify, manifest
python -m explorer.cli external --elliptic      # DS3, with SHA-256; needs a Kaggle
    token or prints steps
python -m explorer.cli external --bitcoinheist # DS4, with SHA-256
python -m lm.prepare_text --delete-raw          # existing: DS5, tokenizer + 2B train +
    10M val tokens
python -m explorer.cli corpus --target 5000     # DS2 overnight; rerun the same command
    to resume
python -m lm.prepare_chat                       # existing: DS6 tokens and loss masks
```

D.3 GPU lane (Sections 5.3 and 10)

```
python -m lm.train --config configs/text_150m_pretrain.json --bench   # existing:
    tokens/s, memory
#   apply the gate: edit total_tokens in the config if 2B tokens would take over 72 h
python -m lm.train --config configs/text_150m_pretrain.json           # existing: Ctrl+
    C pauses,
#   the same command resumes from runs/text-150m/ckpt_last.pt
python -m lm.prepare_tx                         # DS2 -> train/val/test.bin + meta.json
    (bucket edges)
python -m lm.train --config configs/txgpt_m.json --seed 1             # one of 9 runs (
    S, M, L x seeds 1-3);
#   --seed is a small new option for lm.train
python -m lm.tx_baselines                       # global, address-habit and Markov NLL
    on the test split
python -m lm.tx_score --model runs/txgpt-m-s1/model_best.pt           # surprisal for
    every DS1 transaction
```

```
python -m lm.train --config configs/text_150m_chat.json              # existing: chat
    fine-tune
python -m explorer.cli build-ds7               # DS7 pairs, split by source address
python -m lm.prepare_chat --source jsonl --jsonl data/processed/ds7/train.jsonl --out
    data/ds7
python -m lm.train --config configs/text_150m_explain.json           # explanation
    fine-tune
python -m explorer.cli assistant-eval          # faithfulness, limitation line, banned
    words, ROUGE-L
```

D.4 Processing (Section 7)

```
python -m explorer.cli transactions            # clean table + cleaning log
python -m explorer.cli features                # F1-F8, z-scores, scorable flag (
    refined method)
python -m explorer.cli features --as-written   # A1 method as written, for the ablation
python -m explorer.cli split                   # fit/evaluation periods per address
python -m explorer.cli inject                  # tuning plants (seeds 40000-40004),
    reporting plants (30049-30053)
python -m explorer.cli process-external        # DS3 join, degrees, temporal split; DS4
     transforms and sampling
```

D.5 Analysis and models (Sections 8 and 9)

```
jupyter lab notebooks/01_eda.ipynb             # analysis; calls explorer.plots only
python -m explorer.cli baseline --sweep        # thresholds 2.5-5.0 -> tuning_baseline.
    csv
python -m explorer.cli outliers                # Isolation Forest and LOF searches ->
    tuning_outliers.csv
python -m explorer.cli clustering              # K-Means k=2..10, DBSCAN grid, address
    profiles
python -m explorer.cli regression              # naive, Ridge, random forest, gradient
    boosting, quantiles
python -m explorer.cli classify --dataset elliptic --n-jobs 12        # overnight
python -m explorer.cli classify --dataset heist --splits temporal,random,lofo
python -m explorer.cli label-check             # unsupervised scores against DS3/DS4
    labels
```

D.6 Evaluation and outputs (Section 11)

```
python -m explorer.cli evaluate                # every table in reports/tables/ (CSV + .
    tex)
python -m explorer.cli reviewer-sheet          # 40 shuffled rows, no flag column
python -m explorer.cli reviewer-score --ratings ratings_mkn.csv ratings_cdb.csv   #
    agreement, kappa
python -m explorer.cli figures                 # vector PDFs in reports/figures/
python -m explorer.cli all --offline           # everything above from the shipped data
     (README step 4)
```

D.7 Prediction (Section 12.1)

```
python -m explorer.cli predict address bc1q...
python -m explorer.cli predict what-if --address bc1q... --amount 0.05 --direction out
    --time 2026-10-01T12:00Z
python -m explorer.cli predict elliptic --rows data/processed/elliptic/example_rows.csv
python -m explorer.cli predict heist --length 18 --weight 0.008 --count 1 --looped 0 --
    neighbors 2 --income 1.2e8
```

```
python -m lm.chat --model models/assistant     # existing chat loop, on the final
    weights
```

D.8 Report checks and packaging (Sections 13 and 15.4)

```
python tools/wordcount.py report/              # body words, Introduction to Conclusion
python tools/refcheck.py report/               # cited-but-unlisted and listed-but-
    uncited: both empty
python tools/numbercheck.py report/ reports/tables/
python tools/bannedwords.py report/ explorer/ README.md
pdflatex main && pdflatex main                 # in report/, twice for references
git tag v1.0
python tools/package.py --tag v1.0             # zip + SHA256SUMS.txt, from the tag
    only
certutil -hashfile HCMC_COS30049.1_4_Assignment2_Code.zip SHA256   # Windows check of
    the zip itself
```